

# The JETSPIN User Manual

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## About JETSPIN

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If results obtained with this code are published, an appropriate citation would be:

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# Chapter 1

## Introduction

### Scope of Chapter

This chapter describes the design and directory structure of JETSPIN.

### 1.1 The JETSPIN Package

In the recent years, electrospun nanofibers have gained a considerable industrial interest due to many possible applications, such as tissue engineering, air and water filtration, drug delivery and regenerative medicine. In particular, the high surface-area ratio of the fibers offers an intriguing prospect for technological applications. As consequence, several studies were focused on the characterization and production of uni-dimensionally elongated nanostructures. A number of reviews [1, 2, 3, 4, 5] and books [6, 7, 8] concerning electrospinning have been published in the last decade.

Typically, electrospun nanofibers are produced at laboratory scale by the uniaxial stretching of a jet, which is ejected at the nozzle from a charged polymer solution. The initial elongation of a jet can be produced by applying an externally electrostatic field between the spinneret and a conductive collector. Electrospinning involves mainly two sequential stages in the uniaxial elongation of the extruded polymer jet: an initial quasi steady stage, in which the electric field stretches the jet in a straight path away from the nozzle of the ejecting apparatus, and a second stage characterized by a bending instability induced from small perturbations, which misalign the jet from its axis of elongation [9]. These small disturbances may originate from mechanical vibrations at the nozzle or from hydrodynamic-aerodynamic perturbations within the experimental apparatus. Such a misalignment provides an electrostatic-driven bending instability before the jet reaches the conductive collector, where the fibers are finally deposited. As a consequence, the jet path length between the nozzle and the collector increases, and the stream cross-section undergoes a further decrease. The ultimate goal of electrospinning process is to minimize the radius of the collected fibers. By a simple argument of mass conservation, this is tantamount to maximizing the jet length by the time it reaches the collecting plane. By the same argument, it is therefore of interest to minimize the length of the initial stable jet region. Consequently, the bending instability is a desirable effect, as it produces a higher surface-area-to-volume ratio of the jet, which is transferred to the resulting nanofibers [10].

Computational models provide a useful tool to elucidate the physical phenomenon and provide information which might be used for the design of electrospinning experiments. Numerical simulations can be used to improve the capability of predicting the key processing parameters and exert a better control on the resulting nanofiber structure. In recent years, with a renewed interest in nanotechnology, electrospinning studies attracted the attention of many researchers both from modeling, computational and experimental point of view [11, 12, 13, 14, 15, 9]. Although some author uses dissipative particle dynamics (DPD) mesoscale simulation method into electrospinning study [16], most models usually treat the jet filament as a series of discrete elements (*beads*) obeying the equations of continuum mechanics [11, 12]. Each bead is subject to different types of interactions, such as long-range Coulomb repulsion, viscoelastic drag, the external electric field. The main aim of such models is to explain the complexity of the resulting dynamics and provide the

set of parameters driving the optimal process. The effect of fast-oscillating loads on the bending instability have been explored in an extensive modeling and computational study [17].

JETSPIN delivers a FORTRAN code especially designed to simulate the electrospinning process in a variety of different conditions and experimental settings. This comprehensive platform is devised such that different cases and input variables can be described and simulated. The framework is developed to exploit several computational architectures, both serial and parallel.

JETSPIN, as open-source software, can be used to carry out a systematic sensitivity analysis over a broad range of parameter values. The results of simulations provide valuable insight on the physics of the process and can be used to assess experimental procedures for an optimal design of the equipment and to control processing strategies of technological advanced nanofibers.

**The JETSPIN code is freely downloadable at the web site:**

<http://www.nanojets.eu>

## 1.2 Structure

JETSPIN is written in free format FORTRAN90, and it consists of approximately 160 subroutines. The source exploits the modular approach provided by the programming language. All the variables having in common description of certain features or method are grouped in modules. The convention of explicit type declaration is adopted, and all the arguments passed in calling sequences of functions or subroutines have defined intent. We use the PRIVATE and PUBLIC accessibility attributes in order to decrease error-proneness in programming.

The main routines have been gathered in the `main.f90` file, which drives all the CPU-intensive computations needed for the capabilities mentioned below. The variables describing the main features of nanofibers (position, velocity, etc.) are declared in the `nanojet_mod.f90` file, which also contains the main subroutines for the memory management of the fundamental data of the simulated system. Since the size system is strictly time-dependent, JETSPIN exploits the dynamic array allocation features of FORTRAN90 to assign the necessary array dimensions. In particular, the size system is modified by the routines `add_jetbead` and `erase_jetbead`, while the decision of the main array size, declared as `mxnpjet`, is handled by the routine `reallocate_jet`. The sizes of various service bookkeeping arrays are handled within a parallel implementation strategy, which exploits few dedicated subroutines (see Sec 1.3). All the implemented time integrators are written in the `integrator_mod.f90` file, which contains the routine `driver_integrator` to select the proper integrator, as indicated in the input file. All the terms of equations of motion for the implemented model (see Sec 2.1) are computed by routines located in the `eom_mod.f90` file, which call other subroutines in the files `coulomb_force_mod.f90`, `viscoelastic_force_mod.f90` and `support_functions_mod.f90`. A summarizing scheme of the main JETSPIN program in the `main.f90` file has been sketched in Fig 1.1.

The user can carry out simulations of nanofibers without a detailed understanding of the structure of JETSPIN code. All the parameters governing the system can be defined in the input file (see Sec 3.1), which is read by routines located in the `io_mod.f90` and `parse_mod.f90` files. Instead, the user should be acquainted with the model described in Cap 2. The content of the output file is completely customizable by the input file as described in Sec 3.1, and it can report different time-averaged observables computed by routines of the module `statistic_mod` (see Sec 3.2). The routines in the `error_mod.f90` file can display various warning or error banners on computer terminal, so that the user can easily correct the most common mistakes in the input file.

JETSPIN is supplied as a single UNIX compressed (tarred and gzipped) directory with four sub-directories. All the source code files are contained in the *source* sub-directory. The *examples* sub-directory contains different test cases that can help the user to edit new input files. The *build* sub-directory stores a UNIX `makefile` that assembles the executable versions of the code both in serial and parallel versions with different compilers. It is invoked from the repository root while using the *source* sub-directory as its working directory (see Section 1.4); it does not need to be copied. Finally, after JETSPIN was compiled, the *execute* sub-directory contains the binary executable file, where it can be run.

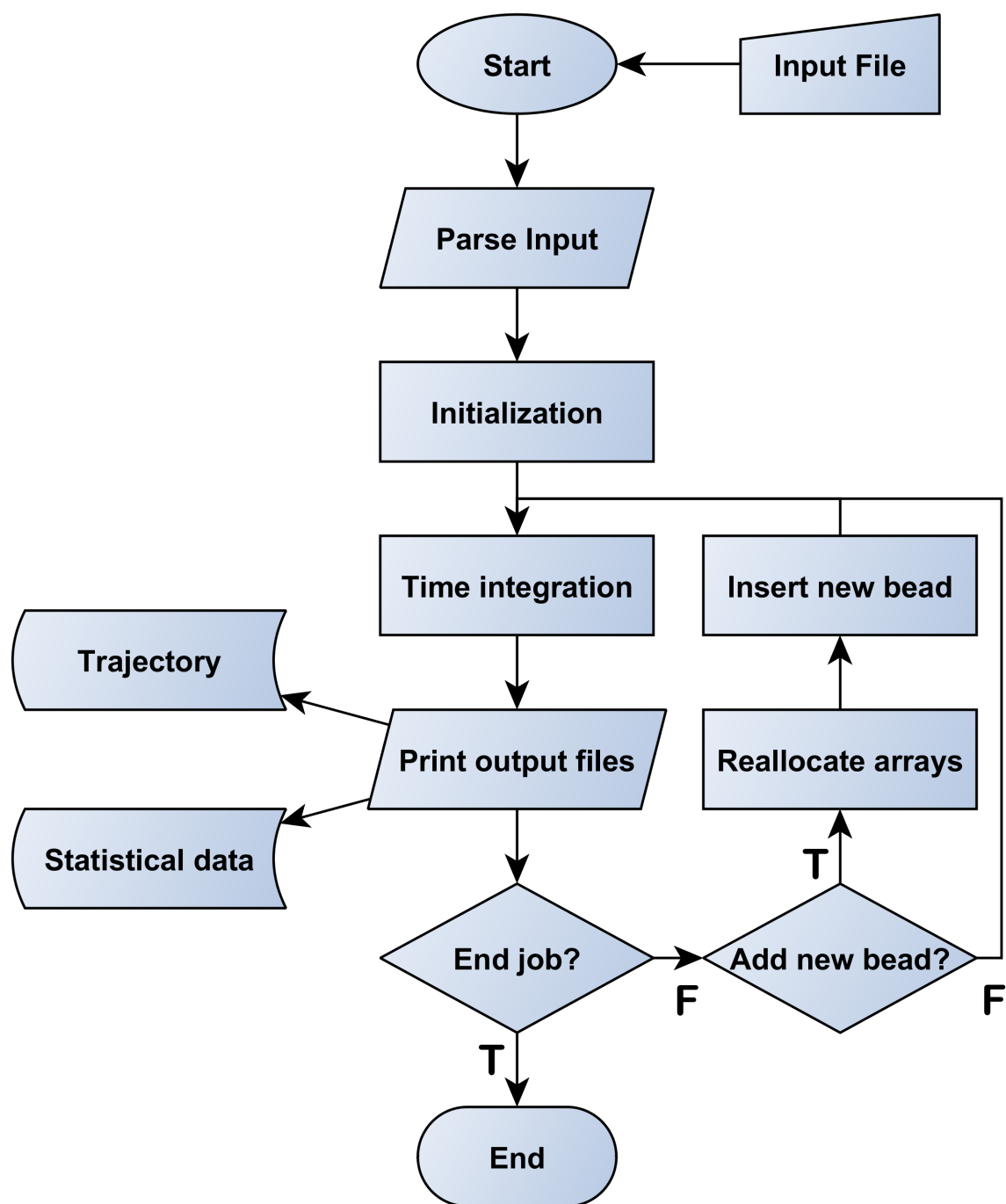


Figure 1.1: Structure of the main JETSPIN program.



### 1.3 Parallelization

The parallel infrastructure of JETSPIN incorporates the necessary data distribution and communication structures. The parallel strategy underlying JETSPIN is the Replicated Data (RD) scheme[18], where fundamental data of the simulated system are reproduced on all processing nodes. In simulations of nanofibers, the fundamental data consist of position, velocity, and viscoelastic stress arrays of each bead in which the nanojet is discretised (see below Cap 2). Further data defining mass and charge of each bead are also replicated. However, all auxiliary data are distributed in equal portion of data (as much as possible) for each processor. Despite being available other parallel strategies such as the Domain Decomposition[19], our experience has shown that such volume of data is by no means prohibitive on current parallel computers.

By the RD scheme, we adopt in JETSPIN a simple communication strategy of the communication between nodes, which is handled by global summation routines. The module `version_mod` located in the `parallel_version_mod.f90` file contains all the global communication routines which exploit the MPI (Message Passing Interface) library. Note that a FORTRAN90 compiler and an MPI implementation for the specific machine architecture are required in order to compile JETSPIN in parallel mode. An alternative version of the module `version_mod` is located in the `serial_version_mod.f90` file, and it can be easily selected by appropriate flags in the `makefile` at compile time. By selecting this version, JETSPIN can also be run on serial computers without modification, even though the code has been designed to run on parallel computers.

The size system is strictly time-dependent as mentioned in Sec 1.2 and, therefore, the memory of various service bookkeeping arrays is dynamically distributed over all the processing nodes. In particular, the bookkeeping array size, declared as `mxchunk`, is managed by the routine `set_mxchunk`. All the beads of the discretised nanofiber are assigned at every time step to a specific node by the routine `set_chunk`, and their temporary data are stored in bookkeeping arrays belonging to the assigned node. It is worth stressing that the communication latency makes the parallelization efficiency strictly dependent on the system size. Therefore, we only advice the use of JETSPIN in parallel mode whenever the user expects a system size with at least 50 beads for each node (further details in Ref [20]).

### 1.4 Compiling the Source Code

The *build* sub-directory stores a UNIX **Makefile** that assembles serial and parallel executable versions with different compilers. JETSPIN may be compiled on any UNIX platform. From the root of the repository, invoke this file directly while making *source* the working directory; there is no need to copy the Makefile. Use

```
make -C source -f ../build/Makefile target
```

where `target` is one of the options reported in Table 1.1. Object and module files are generated in *source*, and the resulting `main.x` executable is placed in *execute*. For example, the serial GFortran build is produced with

```
make -C source -f ../build/Makefile gfortran
```

and the corresponding Open MPI build with target `gfortran-mpi`. Build products can be removed with target `clean` using the same command form.

NVIDIA HPC SDK provides separate CPU-only and OpenACC build targets. On a module-based installation, first load the SDK and verify that its MPI wrapper uses NVFORTRAN:

```
module purge
module use /opt/nvidia/hpc_sdk/modulefiles
module load nvhpc/24.3
nvfortran --version
mpif90 --showme:command
```

The serial and MPI CPU-only builds are selected respectively with `nvfortran` and `nvfortran-mpi`. The latter requires `mpif90 -showme:command` to report `nvfortran`. Neither target enables OpenACC offloading.

The `nvfortran-openacc` target enables single-GPU OpenACC offloading. Its default values target compute capability 8.0, including NVIDIA A30 and A100 GPUs, and the CUDA 12.3 toolkit distributed with HPC SDK 24.3. It also selects the `nofma` GPU option to preserve the numerical behaviour of the nearly straight three-point curvature calculation:

```
make -C source -f ../build/Makefile nvfortran-openacc
```

The OpenACC GPU and host-validation targets expose the standard `_OPENACC` preprocessing macro. Accelerator imports, initialization, Coulomb dispatch, and equation-of-motion dispatch are enclosed by this macro. CPU targets preprocess the same sources without defining it, so accelerator calls are absent from their compiled translation units rather than being retained as no-op runtime branches.

Another device architecture or installed CUDA toolkit can be selected through the `GPUCC` and `CUDA_VERSION` Make variables. The compute capability is passed without the `cc` prefix; for example:

```
make -C source -f ../build/Makefile nvfortran-openacc \
  GPUCC=90 CUDA_VERSION=12.3
```

Target `nvfortran-openacc-host` compiles the OpenACC control path for host execution. It is intended for compiler and numerical validation when an NVIDIA device is unavailable and is not a GPU performance build. Before a device run, `nvidia-smi` and `nvaccelinfo` should both confirm that the selected GPU is visible.

On Windows system we advise the user to compile JETSPIN under the command-line interface Cygwin [21]. Finally, the binary executable file can be run in the *execute* sub-directory. Note that a FORTRAN 90 compiler is necessary in order to compile JETSPIN. Further, a Message Passing Interface Implementation should be also present, if the user wishes to compile JETSPIN in parallel mode. We recommend GFORTRAN as FORTRAN 90 compiler, but IFORT-Intel(R) Fortran Compiler or G95 can also be used. As Message Passing Interface Implementation, we suggest to use that provided by the Open MPI Project.

| target:                | meaning:                                                                                                                                                                                                                |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| gfortran               | compile in serial mode using the GFortran compiler.                                                                                                                                                                     |
| gfortran-mpi           | compile in parallel mode using the GFortran compiler and the Open Mpi library.                                                                                                                                          |
| cygwin                 | compile in serial mode using the GFortran compiler under the command-line interface Cygwin for Windows.                                                                                                                 |
| cygwin-mpi             | compile in parallel mode using the GFortran compiler and the Open Mpi library under the command-line interface Cygwin for Windows (note a pre-compiled package of the Open Mpi library is already available on Cygwin). |
| intel                  | compile in serial mode using the Intel compiler.                                                                                                                                                                        |
| intel-mpi              | compile in parallel mode using the Intel compiler and the Intel Mpi library.                                                                                                                                            |
| intel-openmpi          | compile in parallel mode using the Intel compiler and the Open Mpi library.                                                                                                                                             |
| nvfortran              | compile in serial CPU mode using NVIDIA Fortran; OpenACC offloading is disabled.                                                                                                                                        |
| nvfortran-mpi          | compile in parallel CPU mode using an MPI wrapper configured with NVIDIA Fortran; OpenACC offloading is disabled.                                                                                                       |
| nvfortran-openacc      | compile a single-GPU OpenACC executable; configure the device with GPUCC and CUDA_VERSION.                                                                                                                              |
| nvfortran-openacc-host | compile the OpenACC code path for host-only validation.                                                                                                                                                                 |
| help                   | return the list of possible target choices                                                                                                                                                                              |

Table 1.1: List of targets for several common workstations and parallel computers, which can be used with the Makefile in the *build* sub-directory.

## 1.5 Running JETSPIN

In order to run the JETSPIN executable (*main.x*) the user should prepare the input file following the hints provided in Section 3.1. The test cases located in *examples* sub-directory can be used as starting point for editing the input file. As JETSPIN is executed, the program will look for the input file, named *input.dat*, in the same directory, where the executable was called. For instance, if JETSPIN is called in the *execute* sub-directory, the code will look for the file *input.dat*, which must have been put in the same sub-directory.

The JETSPIN code can be executed in serial mode by the command

```
./main.x
```

or in parallel mode by the command

```
mpirun -np $ncpu ./main.x
```

where *\$ncpu* is the number of CPUs used in parallel (e.g. *mpirun -np 4 ./main.x* in order to run JETSPIN in parallel mode on 4 CPUs).

A successful run of JETSPIN will generate several output files, which appear in the directory, where the executable was called. The output files will be produced only if their writing was explicitly requested in input file by using proper directives (see Section 3.1). A detailed description of the generated output files is provided in Section 3.2. Briefly, the main output files are the files *statdat.dat*, *traj.xyz* and *frame%06d.xyz*, which are human readable. The contains a catalogue of instantaneous or blocked averaged values of several variables, which can be used for temporal or statistical analysis in post-processing. The *traj.xyz* file provides a time ordered sequence of configurations to facilitate further analysis of the jet dynamics, while the *frame%06d.xyz* file, reports the same information in splitted files, one for each registered time frame. Further, as JETSPIN

is running, a series of lines will be printed in the terminal window, providing several information as requested by suitable directives in input file (see Section 3.1). Also present will be the `save.dat` file, which contains the accumulated data for computing statistical quantities and other quantities defining the current simulation. This file is intended to be used together with the input file for restarting a JETSPIN job (see Section 1.6), and it is *not* human readable.

## 1.6 Restarting JETSPIN

A simulation can be restarted after normal termination or error termination, if the last was due to inadequate time allocation. In order to restart a JETSPIN job the file `input.dat` is again required together with the file `restart.dat`, which is the exact copy of the `save.dat` file created by the previous job. In particular, the `save.dat` file was written by JETSPIN at regular time interval during the previous simulation. The file `save.dat` is *not* human readable, and it should be renamed as `restart.dat`, and located in the same directory of input file. If the user attempts to restart JETSPIN without the `restart.dat` file, the job will immediately exit with error. Note that the directive “*restart yes*” needs to be added in input file in order to restart a previous JETSPIN job (see Section 3.1). Further, whenever a job is restarted, it is also possible to reset the accumulated statistical data, collected in the previous run, by adding the directive “*restart reset*” in input file. It is worth stressing that JETSPIN will append new data to the files `statdat.dat`, `traj.xyz`, while all the other output files will be **overwritten** (included the `save.dat` file). Thus, the user is advised to keep backup copies of the `restart.dat` file, noting the times it was written, to help you avoid going right back to the start of a simulation.

The same restarting procedure can be also used for extending a simulation beyond its initial allocation of timesteps. In this case, the directive *timesteps* should be adjusted in input file to the new total number of steps required for the simulation. For instance, if a 10000 step simulation needs to be extended by a further 5000 steps, the user should use the directive “*timesteps 15000*” in the input file and include the “*restart yes*” directive. Further, the restarting procedure can be useful in the event of machine failure. Here, the job can be restarted in the same way from the `save.dat` file, which is written at regular time intervals in order to meet just such an emergency. We stress that the procedure is not foolproof in the last event, since the job could be crash while the `save.dat` file is being written.

## Chapter 2

# JETSPIN Model and Algorithms

### Scope of Chapter

This chapter describes the model and simulation algorithms incorporated into JETSPIN.

### 2.1 Equations of Motion

The model implemented in JETSPIN is an extension of the Lagrangian discrete model introduced by Reneker et al. [11] (in the following text referred as Ren model). The model provides a compromise of efficiency and accuracy by representing the filament as a series of  $n$  beads (jet beads) at mutual distance  $l$  connected by viscoelastic elements. The length  $l$  is typically larger than the cross-sectional radius of the filament, but smaller than the characteristic lengths of other observables of interest (e.g. curvature radius). Each  $i$ -th bead has mass  $m_i$  and charge  $q_i$  (not necessarily equal for all the beads). The nanofiber is modeled as a viscoelastic fluid following the approach developed in Ref. [22], and the stress  $\sigma_i$  for the element connecting bead  $i$  with bead  $i + 1$  is given by the viscoelastic constitutive equation:

$$\frac{d\sigma_i}{dt} = -\frac{1}{t_0} (\sigma_i - \sigma_{HB}), \quad (2.1)$$

where  $t_0$  is the relaxation time and  $\sigma_{HB}$  is the Herschel–Bulkley stress. Note that the relaxation time is given as

$$t_0 = \frac{\mu}{G}, \quad (2.2)$$

where  $G$  is the elastic modulus and  $\mu$  the viscosity of the fluid jet, while  $\sigma_{HB}$  reads

$$\sigma_{HB} = \sigma_Y + K_{ve} \left( \frac{1}{l_i} \frac{dl_i}{dt} \right)^{n_{ve}}. \quad (2.3)$$

In the previous expression,  $K_{ve}$  is the flow consistency index with dimensions  $\text{gs}^{ne-2}\text{cm}^{-1}$ ,  $\sigma_Y$  the yield stress,  $n_{ve}$  the flow behavior index,  $t$  the time, and  $l_i$  the length of the element connecting bead  $i$  with bead  $i + 1$ . Note that it is possible to define an effective viscosity  $\mu_{eff}$  as

$$\mu_{eff} = K_{ve} \left( \frac{1}{l_i} \frac{dl_i}{dt} \right)^{n_{ve}-1}. \quad (2.4)$$

It is worth stressing that the case  $K_{ve} = \mu$ ,  $n_{ve} = 1$  and  $\sigma_Y = 0$  recovers the Maxwellian fluid model. Keeping the last case in mind, for convenience sake, we write

$$K_{ve} = k_{ve}\mu, \quad (2.5)$$

with  $k_{ve}$  the flow consistency index re-scaled by  $\mu$ , and the viscoelastic constitutive equation is rearranged as

$$\frac{d\sigma_i}{dt} = -\frac{G}{\mu} \left[ \sigma_i - \sigma_Y + k_{ve}\mu \left( \frac{1}{l_i} \frac{dl_i}{dt} \right)^{n_{ve}} \right], \quad (2.6)$$

The last constitutive equation allows the modeling of several fluids as, for example, Bingham fluids (  $\sigma_Y > 0$  ), shear-thinning fluids (  $n_{ve} > 1$  ), shear-thickening fluid (  $n_{ve} < 1$  ).

As alternative, it is possible to activate the Kelvin–Voigt model [23] (see Tab 3.1). In the last case, Eq 2.6 is replaced by

$$\frac{d\sigma_i}{dt} = G \frac{1}{l_i} \frac{dl_i}{dt} + \mu \frac{1}{l_i} \frac{dv_i}{dt}. \quad (2.7)$$

Given  $a_i$  the cross-sectional radius of the filament at the bead  $i$ , the viscoelastic force  $\vec{f}_{ve}$  pulling bead  $i$  back to  $i - 1$  and towards  $i + 1$  is

$$\vec{f}_{ve,i} = -\pi a_i^2 \sigma_i \cdot \vec{t}_i + \pi a_{i+1}^2 \sigma_{i+1} \cdot \vec{t}_{i+1}, \quad (2.8)$$

where  $\vec{t}_i$  is the unit vector pointing bead  $i$  from bead  $i - 1$ . The surface tension force  $\vec{f}_{st}$  for the  $i - th$  bead is given by

$$\vec{f}_{st,i} = k_i \cdot \pi \left( \frac{a_i + a_{i-1}}{2} \right)^2 \alpha \cdot \vec{c}_i, \quad (2.9)$$

where  $\alpha$  is the surface tension coefficient,  $k_i$  is the local curvature, and  $\vec{c}_i$  is the unit vector pointing the center of the local curvature from bead  $i$ . Note the force  $\vec{f}_{st}$  is acting to restore the rectilinear shape of the bending part of the jet.

In the electrospinning experimental configuration an intense external electric potential  $\Phi_0$  is applied between the spinneret and a conducting collector located at distance  $h$  from the injection point. As consequence, each  $i - th$  bead endures the electric force

$$\vec{f}_{el,i} = e_i \frac{\Phi_0}{h} \cdot \vec{x}, \quad (2.10)$$

where  $\vec{x}$  is the unit vector pointing the collector from the spinneret and assumed along the  $x$  axis. Note that in Ren model the electric field  $\Phi_0$  is assumed to be static in order to avoid the computationally expensive integration of Poisson equation, whereas in reality  $\Phi$  is depending on the net charge of the nanofiber so as to maintain constant the potential at the electrodes. The latter issue was elegantly addressed by Kowalewski et al. [24], and its implementation in JETSPIN will be planned. Further, it is possible to impose a time dependent external electric potential described by a rectangular waveform with amplitude  $\Phi_0$  and frequency given in input file (see Section 3.1).

The net Coulomb force  $\vec{f}_c$  acting on the  $i - th$  bead from all the other beads is given by

$$\vec{f}_{c,i} = \sum_{\substack{j=1 \\ j \neq i}}^n \frac{q_i q_j}{R_{ij}^2 + a_j^2} \cdot \vec{u}_{ij}, \quad (2.11)$$

where  $R_{ij} = \left[ (x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2 \right]^{1/2}$ , and  $\vec{u}_{ij}$  is the unit vector pointing the  $i - th$  bead from  $j - th$  bead. Note that  $j - th$  charge is assumed to induces a field from its outer shell of the fiber (a charged ring of radius  $a_j$ ) and not from the center line (the particle-like point) as implemented in the original Ren model. Although the Ren model provides a reasonable description for the spiral motion of the jet, the last term  $\vec{f}_c$  introduces mathematical inconsistencies due to the discretization of the fiber into point-charges. Different approaches were developed to overcome this issue which usually imply strong approximations[25, 10, 26]. Other strategies use a less crude approximation by accounting for the actual electrostatic form factors between two interacting sections of a charged fiber[27] or involving more sophisticated methods which exploit the tree-code hierarchical force calculation algorithm[24]. In JETSPIN it is possible to activate the dynamic refinement algorithm to recover a finer discretization of the jet (see Sec 2.4).

Although usually much smaller than the other driving forces, the body force due to the gravity is computed in the model by the usual expression

$$\vec{\mathbf{f}}_{g,i} = m_i g \cdot \vec{\mathbf{x}}, \quad (2.12)$$

where  $g$  is the gravitational acceleration.

As shown in experimental evidences [28], the air drag force affects the dynamics of the nanofiber. The extended stochastic model developed in Refs [29, 30, 31] was included in JETSPIN to reproduce the aerodynamic effects. In particular, the air drag is modeled by adding two force terms: a random term and a dissipative term. The dissipative air drag term is usually dependent on the geometry of the jet, which changes in time, and it combines longitudinal and lateral components. Based on experimental results [28], the longitudinal component of the air drag dissipative force term acting on bead  $i$  is

$$\vec{\mathbf{f}}_{air,i} = -0.65\pi a_i l_i \rho_a (2a_i/\nu_a)^{-0.81} v_{tan,i}^{1+0.19} \cdot \vec{\mathbf{t}}_{i-1} \quad (2.13)$$

where  $\rho_a$  and  $\nu_a$  are the air density and kinematic viscosity, respectively,  $\vec{\mathbf{t}}_{i-1}$  is the unit vector pointing bead  $i$  from bead  $i-1$ , and  $v_{tan,i} = \vec{\mathbf{v}}_i \cdot \vec{\mathbf{t}}_{i-1}$  is the projection of the vector velocity of bead  $i$  along the unit vector  $\vec{\mathbf{t}}_{i-1}$ . Further, it is possible to write  $\vec{\mathbf{f}}_{air,i}$  as

$$\vec{\mathbf{f}}_{air,i} = -m_i \gamma_i l_i^{0.905} v_{t,i}^{1+0.19} \cdot \vec{\mathbf{t}}_{i-1} \quad (2.14)$$

if we introduced the dissipative coefficient  $\gamma_i$  equal to

$$\gamma_i = 0.65\pi^{0.905} \frac{\rho_a}{m_i} \left( \frac{2}{\nu_{air}} \right)^{-0.81} V_i^{0.095}, \quad (2.15)$$

where  $V_i = \pi a_i^2 l_i$  is the jet volume represented by the  $i$ -th bead.

Following the expression introduced by A. Yarin [32, 6], under a high-speed air drag the lateral component  $\vec{\mathbf{f}}_{lft,i}$  of the aerodynamic dissipative force related to the flow speed is given in the linear approximation (for small bending perturbations) by

$$\vec{\mathbf{f}}_{lft,i} = -l_i \cdot k_i \rho_a v_i^2 \pi \left( \frac{a_i + a_{i-1}}{2} \right)^2 \cdot \vec{\mathbf{c}}_i, \quad (2.16)$$

where  $k_i$  is still the local curvature, and  $\vec{\mathbf{c}}_i$  is the unit vector pointing the center of the local curvature from bead  $i$ . The combined action of such longitudinal and lateral components provide the dissipative force term acting on the  $i$ -th bead

$$\vec{\mathbf{f}}_{diss,i} = \vec{\mathbf{f}}_{air,i} + \vec{\mathbf{f}}_{lft,i}. \quad (2.17)$$

Instead, the random force term for the  $i$ -th bead has the form

$$\vec{\mathbf{f}}_{rand,i} = \sqrt{2m_i^2 D_v} \cdot \vec{\boldsymbol{\eta}}_i(t), \quad (2.18)$$

where  $D_v$  denotes a generic diffusion coefficient in velocity space (which is assumed constant and equal for all the beads), and  $\vec{\boldsymbol{\eta}}_i$  is a three-dimensional vector, whereof each component  $\eta$  is an independent stochastic process that is nowhere differentiable with  $\langle \eta(t_1) \eta(t_2) \rangle = \delta(|t_2 - t_1|)$ , and  $\langle \eta(t) \rangle = 0$ . Note that for the sake of simplicity we assume  $\eta = d\omega(t)/dt$ , where  $\omega(t)$  is a Wiener process. Further details are available in Ref [33].

The combined action of these forces governs the elongation of the jet according to the Newton's equation providing a non-linear Langevin-like stochastic differential equation:

$$m_i \frac{d\vec{\mathbf{v}}_i}{dt} = \vec{\mathbf{f}}_{el,i} + \vec{\mathbf{f}}_{c,i} + \vec{\mathbf{f}}_{ve,i} + \vec{\mathbf{f}}_{st,i} + \vec{\mathbf{f}}_{g,i} + \vec{\mathbf{f}}_{diss,i} + \vec{\mathbf{f}}_{rand}, \quad (2.19)$$

where  $\vec{\mathbf{v}}_i$  is the velocity of the  $i$ -th bead. The velocity  $\vec{\mathbf{v}}_i$  satisfies the kinematic relation:

$$\frac{d\vec{\mathbf{r}}_i}{dt} = \vec{\mathbf{v}}_i \quad (2.20)$$

where  $\vec{r}_i$  is the position vector of the  $i$ -th bead,  $\vec{r}_i = x_i\vec{x} + y_i\vec{y} + z_i\vec{z}$ . The three Eqs 2.6, 2.19 and 2.20 form the set of equations of motion (EOM) governing the time evolution of system. It is worth stressing that Eq 2.19 recovers a deterministic EOM by imposing  $\rho_a = 0$  and  $D_v = 0$ .

## 2.2 Perturbations at the nozzle

The spinneret nozzle is represented by a single mass-less point of charge  $\bar{q}$  fixed at  $x = 0$  (nozzle bead). Its charge  $\bar{q}$  is assumed equal to the mean charge value of the jet beads. Such charged point can be also interpreted as a small portion of jet which is fixed to the nozzle. In JETSPIN it is possible to add small perturbations to the  $y_n$  and  $z_n$  coordinates of the nozzle bead in order to model fast mechanical oscillations of the spinneret[17]. Given the initial position of the nozzle

$$y_n = A \cdot \cos(\varphi) \quad (2.21a)$$

$$z_n = A \cdot \sin(\varphi), \quad (2.21b)$$

the equations of motion for the nozzle bead are

$$\frac{dy_n}{dt} = -\omega \cdot z_n \quad (2.22a)$$

$$\frac{dz_n}{dt} = \omega \cdot y_n, \quad (2.22b)$$

where  $A$  denotes the amplitude of the perturbation, while  $\omega$  and  $\varphi$  are its frequency and initial phase, respectively.

## 2.3 Jet insertion

The jet insertion at the nozzle is modeled as follows. For the sake of simplicity, let us consider a simulation which starts with only two bodies: a single mass-less point fixed at  $x = 0$  representing the spinneret nozzle, and a bead modeling an initial jet segment of mass  $m_i$  and charge  $e_i$  located at distance  $l_{step}$  from the nozzle along the  $x$  axis. Here,  $l_{step}$  denotes the length step used to discretize the jet in a sequence of beads. The starting jet bead is assumed to have a cross-sectional radius  $a_0$ , defined as the radius of the filament at the nozzle before the stretching process. Applying the condition that the volume of the jet is conserved, the relation  $\pi a_i^2 l_i = \pi a_0^2 l_{step}$  is valid for any  $i$ -th bead. Furthermore, the starting jet bead has an initial velocity  $v_s$  along the  $x$  axis equal to the bulk fluid velocity in the syringe needle. Once this traveling jet bead is a distance of  $2 \cdot l_{step}$  away from the nozzle, a new jet bead (third body) is placed at distance  $l_{step}$  from the nozzle along the straight line joining the two previous bodies. Let us now label  $i - 1$  the farthest bead from the nozzle, and  $i$  the last inserted bead. The  $i$ -th bead is inserted with the initial velocity  $v_i = v_s + v_d$ , where  $v_d$  denotes the dragging velocity computed as

$$v_d = \frac{v_{i-1} - v_s}{2}. \quad (2.23)$$

Here, the dragging velocity should be interpreted as an extra term which accounts for the drag effect of the electrospun jet on the last inserted segment. Note that the actual dragging velocity definition was chosen in order to not alter the strain velocity term  $(1/l_{i-1}) \cdot (dl_{i-1}/dt)$  of Eq 2.1 before and after the bead insertion.

## 2.4 Dynamic refinement

In the original Ren model, each bead represent a constant quantity  $V_c$  of jet volume, so that the relationship  $\pi a_i^2 l_i = \pi a_0^2 l_{step} = V_c$  is valid for any  $i$ -th bead. Despite such relationship can be sometimes a useful advantage (as example, see Sec 2.7), the radius reduction  $a_i$  of the electrospun nanofiber provides a significant



increase of the discretization length  $l_i$  value along the jet path (also greater than two order of magnitude). As consequence, it was observed in literature [24] that the jet discretization close the collector becomes rather coarse to model efficiently the nanofiber. To get around this problem the dynamic refinement procedure developed in Ref [29] can be used in JETSPIN in order to maintain the the length of the elements  $l_i$  below a prescribed characteristic length  $\max l_{max}$ . Whenever a bead length  $l_i$  is larger than  $l_{max}$ , the nanofiber is refined by discretizing uniformly the jet at the length step  $l_{step}$  value given in input file. The discretization is performed by Akima spline interpolations [34, 35] of the main quantities describing the jet beads (positions, fiber radius, stress, velocities). In order to perform the interpolation we proceed following three steps. First of all, we introduce the variable  $\lambda \in [0, 1]$  to parametrize the jet path, where  $\lambda = 0$  identifies the nozzle, and  $\lambda = 1$  the jet at the collector. For each bead we compute its respective  $\lambda_i$  value by using the formula

$$\lambda_i = \frac{1}{l_{path}} \sum_{k=1}^i l_k \quad (2.24)$$

where  $l_{path} = \sum_{k=1}^{N_{beads}} l_k$  is the jet path length from the nozzle to the collector, and  $i = 1$  and  $i = N_{beads}$  denote the jet bead at the nozzle and collector, respectively ( $i = 0$  denotes the nozzle). The set of  $\lambda_i$  values represent the mesh used to build the cubic spline. Given a generic quantity  $y$ , the data  $y_i = y(\lambda_i)$  are tabulated and used in order to compute the coefficients of the *Akima spline*, following the algorithm reported in Ref [35]. Then, we enforce a uniform parametrization of the nanofiber by imposing all the lengths of the elements equal to  $l_{step}$ . The new mesh  $\lambda_i^*$  is defined as

$$\lambda_0^* = 0 \quad \lambda_i^* = \frac{i}{N_{beads}^*}, \quad i = 1, 2, \dots, N_{beads}^*. \quad (2.25)$$

where  $N_{beads}^* = l_{path}/l_{step}$  represents the number of jet beads in the new representation. Finally, the new values  $y(\lambda_i^*)$  are computed for any  $i$ -th bead by the *spline interpolation*. The procedure is repeated for positions, fiber radius, stress and velocities of the jet beads in order to provide the respective quantities in the new mesh  $\lambda_i^*$ . It is worth to dressing that the volume  $V_i = \pi a_i^2 l_i$  represented by each bead is not anymore constant, and, consequently,  $\pi a_i^2 l_i \neq \pi a_0^2 l_{step}$ . It is also possible keep few beads in their positions before and after the refinement procedure in order to reduce possible errors introduced by the interpolation procedure. In other words, these anchor beads are maintained as knots of both the old and new meshes (see directive `dynam refin anchor` in Tab 3.1). In the following text we will refer to this emended version of the Ren model as DyRen model. Further details on the implemented dynamic refinement procedure are available in Ref [29].

## 2.5 Random noise

In JETSPIN, it is possible to add a random noise to Eq. 2.19 in order to model the instability at the nozzle triggering the bending motion. This procedure can be used as an alternative to (or in junction with) the strategy presented in Section 2.2.

In particular, collected in  $\vec{f}_{old,i} = \vec{f}_{el,i} + \vec{f}_{c,i} + \vec{f}_{ve,i} + \vec{f}_{st,i} + \vec{f}_{g,i} + \vec{f}_{diss,i} + \vec{f}_{rand}$  all the force terms, we modify Eq. 2.19 as

$$m_i \frac{d\vec{v}_i}{dt} = \vec{f}_{old,i} - m_i \gamma_{rand} \vec{v}_i + \sqrt{2m_i^2 D_{rand}} \vec{\eta}_i(t), \quad (2.26)$$

following a standard Langevin approach, where  $D_{rand}$  is a diffusion coefficient in the velocity space (in cgs units,  $\text{cm}^2/\text{s}^3$ ) and as  $\gamma_{rand}$  a dissipative term ( $\text{s}^{-1}$ ). We highlight that  $D_{rand}$  sets the width of the fluctuations. In particular, it is possible to demonstrate [36] that the variance  $\sigma_{rand}(t)^2$  of the velocity  $\vec{v}_i$  ( $\text{cm}^2/\text{s}^2$ ) due to only the random and dissipative terms at time  $t$ , computed over an ensemble of stochastic trajectories, is equal to  $\sigma_{rand}(t)^2 \equiv \langle [\vec{v}_i - \langle \vec{v}_i \rangle]^2 \rangle = (D_{rand}/\gamma_{rand}) (1 - e^{-2\gamma_{rand}t})$ , and, consequently,  $\lim_{t \rightarrow \infty} \sigma_{rand}(t)^2 = D_{rand}/\gamma_{rand}$ , which is sometimes called fluctuation-dissipation Langevin relation. In `input` file, the user should specified both  $D_{rand}$  and  $\sigma_{rand}^2$ . The  $\gamma_{rand}$  is automatically set to be equal to  $\gamma_{rand} = D_{rand}/\sigma_{rand}^2$  by exploiting the Langevin relation.

## 2.6 External oscillating electric field

Oscillating orthogonal electric field can bias the electrospinning process, particularly the radius of the electrospun fibers. Further, it can be used as a control mechanism to achieve uniform distributions of polymer filaments at the collector. In JETSPIN it is possible to simulate external oscillating electric field. Denoted the electric field

$$\vec{E} = (E_x, E_y, E_z), \quad (2.27)$$

the directive **external potential type** (see Table 3.1) can be used to select the waveform of the oscillation. The user can select three types of waveform: a rectangular wave oriented along x-axis; a rectangular wave produced by a RC circuit; a rotating electric field of a sinusoidal wave. In the first case the electric field is given by

$$\vec{E} = (E_x [\sin \omega t + \varphi], 0, 0), \quad (2.28)$$

where the symbol  $[\cdot]$  denotes the ceiling function. If the waveform is modeled as produced by an RC circuit, the exponential decay  $E_x \exp(-t/RC)$  is added to the previous equation whenever the field is switched off, and the exponential increase  $E_x [1 - \exp(-t/RC)]$  is added whenever the field is switched on. Here,  $RC$  denotes a decay time (seconds). The last case describes an orthogonal rotating electric field. Here the electric field takes the form

$$\vec{E} = (E_x, E_y \cos \omega t + \varphi, E_z \sin \omega t + \varphi). \quad (2.29)$$

Details on the main effects of orthogonal rotating electric fields are available in Ref. [37].

## 2.7 Dimensionless quantities

In JETSPIN all the variables are automatically re-scaled and stored in dimensionless units. In order to adopt a dimensionless form of the equations of motion, we use the dimensionless scaling procedure proposed by Reneker et al.[11]. We define a characteristic length

$$L_0 = \sqrt{\frac{\bar{q}^2}{\pi a_0^2 G}} = l_{step} \sqrt{\frac{\pi a_0^2 \rho_V^2}{G}}, \quad (2.30)$$

where we write the charge  $q$  as  $\pi a_0^2 l_{step} \rho_V$ , denoting  $\rho_V$  the electric volume charge density of the filament. Further, we divide the time  $t$  and the stress  $\sigma$  by their respective characteristic scales reported in Tab 2.1. Applying the condition that the volume of the jet is conserved  $\pi a_i^2 l_i = V_i$  for any  $i$ -th bead, we rewrite the EOM as:

$$\frac{d\vec{r}_i}{dt} = \vec{v}_i \quad (2.31a)$$

$$\frac{d\bar{\sigma}_i}{dt} = - \left[ \bar{\sigma}_i - \bar{\sigma}_Y + \bar{k}_{ve} \left( \frac{1}{\bar{l}_i} \frac{d\bar{l}_i}{dt} \right)^{n_{ve}} \right] \quad (2.31b)$$

$$\begin{aligned} \frac{d\vec{v}_i}{dt} = & \Phi \cdot \vec{x} + \sum_{\substack{j=1 \\ j \neq i}}^n \frac{Q_{ij}}{\bar{R}_{ij}^2 + \bar{a}_j^2} \cdot \vec{u}_{ij} - F_{ve,i} \bar{V}_i \frac{\bar{\sigma}_i}{\bar{l}_i} \cdot \vec{t}_i + F_{ve,i+1} \bar{V}_{i+1} \frac{\bar{\sigma}_{i+1}}{\bar{l}_{i+1}} \cdot \vec{t}_{i+1} \\ & + A_i \frac{\bar{k}_i}{4} \left( \frac{\sqrt{\bar{V}_i}}{\sqrt{\bar{l}_i}} + \frac{\sqrt{\bar{V}_{i-1}}}{\sqrt{\bar{l}_{i-1}}} \right)^2 \cdot \vec{c}_i + F_g \cdot \vec{x} - \Gamma_i \bar{l}_i^{0.905} \bar{v}_{tan,i}^{1+0.19} \cdot \vec{t}_{i-1} - \\ & - \Lambda_i \frac{\bar{k}_i \bar{v}_{tan,i}^2}{4} \left( \frac{\sqrt{\bar{V}_i}}{\sqrt{\bar{l}_i}} + \frac{\sqrt{\bar{V}_{i-1}}}{\sqrt{\bar{l}_{i-1}}} \right)^2 \cdot \vec{c}_i - \Gamma_{rand} \vec{v}_i + \sqrt{2(\Theta_v + \Theta_{rand})} \cdot \vec{\eta}_i(\bar{t}) \end{aligned} \quad (2.31c)$$

where we used the dimensionless derived variables and groups defined in Tab 2.1. It is worth stressing that the viscoelastic and surface tension force terms are slightly different from the dimensionless form provided by Reneker et al. [11], since we are considering the DyRen model presented in Sec 2.4. In particular, we do not use the relation  $\pi a_i^2 l_i = \pi a_0^2 l_{step}$  since it is valid only for the Ren model (see discussion in Sec 2.4). Note that, whenever the Kelvin–Voigt model is activated, Eq 2.31b is replaced by

$$\frac{d\bar{\sigma}_i}{d\bar{t}} = \frac{1}{\bar{l}_i} \frac{d\bar{l}_i}{d\bar{t}} - \frac{1}{\bar{l}_i} \frac{d^2 \bar{l}_i}{d\bar{t}^2}. \quad (2.32)$$

Finally, the dimensionless EOM of the nozzle are given as

$$\frac{d\bar{y}_i}{d\bar{t}} = -K_s \cdot \bar{z}_i \quad (2.33a)$$

$$\frac{d\bar{z}_i}{d\bar{t}} = K_s \cdot \bar{y}_i, \quad (2.33b)$$

with the dimensionless parameter  $K_s$  defined in Tab 2.1.

| Characteristic Scales                                                             |                                                       |
|-----------------------------------------------------------------------------------|-------------------------------------------------------|
| $L_0 = l_{step} \sqrt{\frac{\pi a_0^2 \rho_V^2}{G}}$                              | $t_0 = \frac{\mu}{G}$                                 |
| $\sigma_0 = G$                                                                    |                                                       |
| Dimensionless Quantities                                                          |                                                       |
| $\bar{l}_i = \frac{l_i}{L_0}$                                                     | $\bar{R}_{ij} = \frac{R_{ij}}{L_0}$                   |
| $\bar{\sigma}_i = \frac{\sigma_i}{\sigma_0}$                                      | $\bar{v}_i = v_i \frac{t_0}{L_0}$                     |
| $\bar{k}_i = k_i L_0$                                                             | $\bar{v}_{tan,i} = v_{tan,i} \frac{t_0}{L_0}$         |
| $\bar{V}_i = \frac{V_i}{L_0^3}$                                                   | $\bar{k}_{ve} = \frac{k_{ve}}{\mu \tau^{(n_{ve}-1)}}$ |
| $\bar{a}_i = \frac{a_i}{L_0}$                                                     |                                                       |
| Dimensionless Groups                                                              |                                                       |
| $\Phi_i = \frac{q_i \Phi_0 \mu^2}{m_i h L_0 G^2}$                                 | $Q_{ij} = \frac{q_i q_j \mu^2}{m_i L_0^3 G^2}$        |
| $F_{ve,i} = \frac{L_0 \mu^2}{m_i G}$                                              | $A_i = \frac{\alpha \mu^2}{m_i G^2}$                  |
| $F_g = \frac{g \mu^2}{L_0 G^2}$                                                   | $K_s = \omega \frac{\mu}{G}$                          |
| $H = \frac{h}{L_0}$                                                               | $L_{step} = \frac{l_{step}}{L_0}$                     |
| $\Gamma_i = \gamma_i \cdot t_0 L_0^{0.905} \left( \frac{L_0}{t_0} \right)^{0.19}$ | $\Theta_v = D_v \cdot \frac{t_0^3}{L_0^2}$            |
| $\Gamma_{rand} = \gamma_{rand} \cdot t_0$                                         | $\Theta_{rand} = D_{rand} \cdot \frac{t_0^3}{L_0^2}$  |
| $\Lambda_i = \frac{\rho_a L_0^3}{m_i}$                                            |                                                       |

Table 2.1: Definitions of the characteristic scales, dimensionless derived variables, and groups employed in the text.

## 2.8 Solvent evaporation and solidification

JETSPIN implements the solvent-evaporation and rheological-solidification model introduced by Yarin, Koombhongse and Reneker [38]. The dimensional mass-transfer correlation is

$$Sh = 0.495 Re^{1/3} Sc^{1/2}, \quad Re = \frac{2a_i |\mathbf{v}_i|}{\nu_a}, \quad Sc = \frac{\nu_a}{D_a}, \quad (2.34)$$

where  $a_i$  is the local jet radius,  $\nu_a$  is the kinematic viscosity of air, and  $D_a$  is the diffusion coefficient of solvent vapour in air. For a discrete jet element of length  $l_i$  and volume  $V_i = \pi a_i^2 l_i$ , the corresponding evaporation equation is

$$\frac{dV_i}{dt} = -0.495 D_a Re_i^{1/3} Sc^{1/2} [c_{s,\text{eq}}(T) - c_{s,\infty}] \pi l_i. \quad (2.35)$$

This is the discrete form of Eq. (34) of Yarin et al. [38]; in the discrete model  $l_i$  is the distance between adjoining beads.

The notation used for humidity in Ref. [38] can be ambiguous because  $c_{s,\infty}$  is described there in terms of relative humidity, whereas Eq. (33) gives the saturation-vapour ratio. JETSPIN makes the convention explicit. The input quantity  $RH$  is a relative humidity in the interval  $[0, 1]$ , and the far-field vapour ratio used in Eq. 2.35 is

$$c_{s,\infty} = RH c_{s,\text{eq}}(T). \quad (2.36)$$

If  $c_{s,\text{eq}}(T)$  is not supplied explicitly, JETSPIN evaluates it from the temperature using the water-vapour correlation of Seaver, Galloway and Manuccia used by Yarin et al. If  $D_a$  is not supplied, the same atmospheric-pressure parametrization is used,

$$D_a = 0.211 \left( \frac{T}{273.15 \text{ K}} \right)^{1.94} \text{ cm}^2/\text{s}, \quad (2.37)$$

corresponding to air at one atmosphere. A different diffusivity can be specified explicitly in the input file.

### 2.8.1 JETSPIN nondimensionalization

The evaporation equation is integrated together with the equations of motion in the standard JETSPIN dimensionless variables. With

$$t_0 = \frac{\mu_0}{G_0}, \quad L_0 = \sqrt{\frac{q_0^2}{\pi a_0^2 G_0}}, \quad (2.38)$$

JETSPIN uses

$$\bar{t} = \frac{t}{t_0}, \quad \bar{l}_i = \frac{l_i}{L_0}, \quad \bar{\mathbf{v}}_i = \frac{t_0}{L_0} \mathbf{v}_i, \quad \bar{V}_i = \frac{V_i}{L_0^3}, \quad (2.39)$$

and

$$\bar{\nu}_a = \nu_a \frac{t_0}{L_0^2}, \quad \bar{D}_a = D_a \frac{t_0}{L_0^2}. \quad (2.40)$$

Both  $Re$  and  $Sc$  are invariant under this rescaling. Consequently, Eq. 2.35 becomes

$$\frac{d\bar{V}_i}{d\bar{t}} = -0.495 \bar{D}_a Re_i^{1/3} Sc^{1/2} [c_{s,\text{eq}}(T) - c_{s,\infty}] \pi \bar{l}_i, \quad (2.41)$$

which is the expression evaluated in `eom_ev_mod.f90`. The arrays `jetvl` and `jetve` contain, respectively, the reference volume  $\bar{V}_i^0$  of the material element before solvent loss and its instantaneous volume  $\bar{V}_i$  after evaporation.

Yarin et al. also introduce the evaporation-specific Deborah number

$$De = \frac{\mu_0/G_0}{a_0^2/D_a} = \frac{D_a t_0}{a_0^2} = \bar{D}_a \left( \frac{L_0}{a_0} \right)^2, \quad (2.42)$$

and the group

$$\delta = \frac{L_0 a_0}{t_0 \nu_a}. \quad (2.43)$$

JETSPIN does not need to store  $De$  and  $\delta$  as independent input parameters: it stores  $\bar{D}_a$  and  $\bar{\nu}_a$  and evaluates the local  $Re$  and  $Sc$  directly. Thus the implementation uses the standard JETSPIN  $L_0, t_0$  scaling while remaining equivalent to the evaporation nondimensionalization of Yarin et al.

## 2.8.2 Polymer concentration and rheological solidification

Conservation of polymer mass gives

$$c_{p,i} = c_{p,0} \frac{V_i^0}{V_i}. \quad (2.44)$$

The viscosity is updated according to Eqs. (37) and (41) of Yarin et al.:

$$\frac{\mu_i}{\mu_0} = 10^{B(c_{p,i}^m - c_{p,0}^m)}. \quad (2.45)$$

For the Yarin–Koombhongse–Reneker model Ref. [38] the relaxation time obeys

$$\frac{\theta_i}{\theta_0} = \frac{c_{p,i}}{c_{p,0}}, \quad (2.46)$$

and, since  $G_i = \mu_i/\theta_i$ ,

$$\frac{G_i}{G_0} = \frac{10^{B(c_{p,i}^m - c_{p,0}^m)}}{c_{p,i}/c_{p,0}}. \quad (2.47)$$

JETSPIN retains a generalized exponent  $t_{\text{ev}}$  through  $\theta_i/\theta_0 = (c_{p,i}/c_{p,0})^{t_{\text{ev}}}$ ; setting `evaporation tconstant` 1 gives exactly Eq. 2.46 and is the Yarin-2001 choice. The default value is one.

For Maxwell rheology, with  $\bar{\sigma}_i = \sigma_i/G_0$ , the resulting constitutive equation is

$$\frac{d\bar{\sigma}_i}{dt} = \frac{1}{\theta_i/\theta_0} \left[ \frac{\mu_i}{\mu_0} \frac{1}{\bar{l}_i} \frac{d\bar{l}_i}{dt} - \bar{\sigma}_i \right], \quad (2.48)$$

which is Eq. (39) of Yarin et al. written in the JETSPIN scaling.

## 2.8.3 Kelvin–Voigt extension with concentration-dependent properties

The concentration laws in Eqs. 2.45 and 2.47 describe material properties and can also be combined with the Kelvin–Voigt constitutive model available in JETSPIN. This combination is an extension of the Yarin concentration-dependent rheology; it should not be attributed to Ref. [38], which used a Maxwell fluid.

In dimensionless variables define

$$r_\mu = \frac{\mu_i}{\mu_0}, \quad r_G = \frac{G_i}{G_0} = \frac{r_\mu}{(c_{p,i}/c_{p,0})^{t_{\text{ev}}}}. \quad (2.49)$$

The Kelvin–Voigt constitutive identity is written as

$$\bar{\sigma}_i = r_G \varepsilon_i + r_\mu \dot{\varepsilon}_i, \quad (2.50)$$

where the overdot denotes differentiation with respect to dimensionless time. Differentiating Eq. 2.50 gives

$$\dot{\bar{\sigma}}_i = r_G \dot{\varepsilon}_i + r_\mu \ddot{\varepsilon}_i + \dot{r}_G \varepsilon_i + \dot{r}_\mu \dot{\varepsilon}_i. \quad (2.51)$$

Thus a time-dependent concentration requires the two product-rule terms that are absent when  $G$  and  $\mu$  are constant.

No additional material parameters are introduced. From polymer-mass conservation,

$$\dot{c}_{p,i} = -c_{p,i} \frac{\dot{V}_i}{V_i}, \quad (2.52)$$

and therefore

$$\dot{r}_\mu = r_\mu \ln(10) B m c_{p,i}^{m-1} \dot{c}_{p,i}, \quad (2.53)$$

$$\dot{r}_G = r_G \left[ \ln(10) B m c_{p,i}^{m-1} - \frac{t_{\text{ev}}}{c_{p,i}} \right] \dot{c}_{p,i}. \quad (2.54)$$

JETSPIN reconstructs the instantaneous elastic strain from the current stress state,

$$\varepsilon_i = \frac{\bar{\sigma}_i - r_\mu \dot{\varepsilon}_i}{r_G}, \quad (2.55)$$

so that no extra per-bead state variable is required.

The extension retains the historical JETSPIN Kelvin–Voigt kinematics:  $\dot{\varepsilon}_i = (1/l_i) dl_i/dt$ , while the acceleration term used for  $\ddot{\varepsilon}_i$  is the existing  $(1/l_i) dv_{\parallel,i}/dt$  approximation. Consequently, when  $\dot{c}_p = 0$ , Eq. 2.51 reduces exactly to the pre-existing JETSPIN Kelvin–Voigt differential equation. At the Yarin solidification cutoff,  $\dot{V}_i = \dot{c}_{p,i} = 0$ , so  $r_\mu$  and  $r_G$  remain frozen. Euler, Heun (RK2) and classical RK4 integrations are supported for the coupled Kelvin–Voigt/evaporation model.

### 2.8.4 Compatibility with dynamic refinement

Solvent evaporation can be combined with JETSPIN dynamic refinement for both Maxwell and Kelvin–Voigt rheology. The refinement procedure treats the reference material volume `jetv1` and the instantaneous post-evaporation volume `jetve` independently. Before an Akima remeshing their totals over the refined portion of the jet are stored; after interpolation the two volume fields are reconstructed and each is renormalized to its own pre-refinement total. Thus refinement preserves both the reference material represented by the beads and the current volume after solvent loss.

After remeshing the local polymer concentration continues to be evaluated from Eq. 2.44, using the newly remapped local ratio  $V_i^0/V_i$ . Because the two fields are interpolated separately, the local concentration profile can be slightly smoothed by remeshing even though the corresponding total volumes are conserved. Backup/restore and parallel broadcast operations likewise include the instantaneous volume whenever evaporation is active.

Dynamic refinement may increase the maximum bead capacity during a run. The Kelvin–Voigt/evaporation integrator therefore resizes its Runge–Kutta workspace whenever the refinement changes the required bead or chunk capacity. This keeps the three-way combination of evaporation, Kelvin–Voigt rheology and dynamic refinement valid after a remeshing that grows the discretization.

Evaporation also reduces the mass represented by a material element. JETSPIN therefore uses

$$\frac{m_i}{m_i^0} = \frac{V_i}{V_i^0}, \quad (2.56)$$

in the acceleration terms. In particular, the external-electric, Coulomb, viscoelastic, surface-tension and aerodynamic accelerations are evaluated with the instantaneous mass. This is the discrete counterpart of the modified momentum balance in Eqs. (35a)–(35d) of Yarin et al.

### 2.8.5 Evaporation cutoff

Yarin et al. stop the evaporation calculation when the solvent mass fraction

$$c_{s,i} = 1 - c_{p,i} \quad (2.57)$$

reaches  $c_{s,i} = 0.1$ . Using Eq. 2.44, the equivalent condition on the volume represented by a bead is

$$\frac{V_i}{V_i^0} = \frac{c_{p,0}}{1 - 0.1} = \frac{c_{p,0}}{0.9}. \quad (2.58)$$

The JETSPIN integrators enforce Eq. 2.58 at every intermediate and final integration stage. Once this limit is reached,  $V_i$ ,  $c_{p,i}$ ,  $\mu_i$ ,  $\theta_i$ , and  $G_i$  remain constant, as in the solidification cutoff adopted in Ref. [38].

## 2.9 Integration schemes

In order to integrate the homogeneous differential equations of motion we discretize time as  $t_i = t_0 + i\Delta t$  with  $i = 1, \dots, n_{steps}$ , where  $n_{steps}$  denotes the number of sub-intervals. In JETSPIN three different integration schemes: the first-order accurate Euler scheme, the second-order accurate Heun scheme (sometimes called second-order accurate Runge-Kutta), and the fourth-order accurate Runge-Kutta scheme[39]. The user can select a specific scheme by using appropriate keys in the input file, as described in Sec 3.1. If the airdrag is activated, the explicit strong order scheme by Platen [40, 41, 42] have to be selected, whereof the order of strong convergence was evaluated equal to 1.5.

This scheme avoids the use of derivatives by corresponding finite differences in the same way as Runge-Kutta schemes do for ODEs in a deterministic setting, and it is briefly summarized as follows.

Let us consider a Brownian motion vector process  $\mathbf{X} = \{\mathbf{X}_t, t\}$  of  $d$ -dimensional satisfying the stochastic differential equation

$$\frac{d\mathbf{X}}{dt} = \mathbf{a}(t, X^1, \dots, X^d) + \mathbf{b}d\Omega \quad (2.59)$$

where  $\mathbf{a}$  and  $\mathbf{b}$  are vectors of  $d$ -dimensional usually called drift and diffusion vector coefficients, and  $\Omega(t)$  denotes a Wiener process. Denoted  $Y_t^k$  the approximation for the  $k$ -th component of the vector  $\mathbf{X}$  at time  $t$ , the integrator has the following form:

$$Y_{t+\Delta t}^k = Y_t^k + b^k \Delta\Omega + \frac{1}{2\sqrt{\Delta t}} \left[ a^k(\tilde{\mathbf{Y}}_+) - a^k(\tilde{\mathbf{Y}}_-) \right] \Delta\psi + \frac{1}{4} \left[ a^k(\tilde{\mathbf{Y}}_+) + 2a^k + a^k(\tilde{\mathbf{Y}}_-) \right] \Delta t, \quad (2.60)$$

with the vector supporting values

$$\begin{aligned} \tilde{\mathbf{Y}}_{\pm} &= \mathbf{Y}_t + \mathbf{a}\Delta t \pm \mathbf{b}\sqrt{\Delta t} \\ \tilde{\Phi}_{\pm} &= \tilde{\mathbf{Y}}_{\pm} \pm \mathbf{b}(\tilde{\mathbf{Y}}_+) \sqrt{\Delta t}. \end{aligned} \quad (2.61)$$

Here,  $\Delta\Omega$  and  $\Delta\psi$  indicate normally distributed random variables constructed from two independent  $N(0, 1)$  standard Gaussian distributed random variables ( $U_1, U_2$ ) by means of the following linear transformation:

$$\begin{aligned} \Delta\Omega &= U_1 \sqrt{\Delta t} \\ \Delta\psi &= \frac{1}{2} \Delta t^{3/2} \left( U_1 + \frac{1}{\sqrt{3}} U_2 \right). \end{aligned} \quad (2.62)$$

Note that in JETSPIN the time step  $\Delta t$  is automatically rescaled by the quantity  $\tau$ , in accordance with the mentioned dimensionless scaling convention.

## 2.10 Multiple Timestep Algorithm for Coulomb forces

For simulations involving a large number of beads in the calculation of the long-range Coulomb interactions JETSPIN offers the possibility of using a multiple timestep algorithm which improves the efficiency. The method is based on that described by Streett *et al* in Ref [43] and modified for the special context of electrospinning process.

In the multiple timestep algorithm a primary cutoff  $r_{pcut}$  for the Coulomb pair interactions is given. The cutoff  $r_{pcut}$  is used to define a standard Verlet neighbor list [44]. Long-range Coulomb forces derived from beads in the neighbor list are usually larger than those due to remaining (far) beads not contained in the list. Therefore, it is possible to assess the Coulomb energy if the forces derived from the beads in the neighbor list are calculated at every timestep, while those from the remaining beads are calculated much less frequently, and are merely extrapolated over the interval. JETSPIN handles this procedure as follows: The Verlet neighbor list is updated at regular intervals  $n_{mult}$ . The interval between updates should be usually of the order of  $\sim 50$  timesteps  $n_{step}$ . Whenever the Verlet neighbor list is updated ( $n_{step} = 1$ ), the force contributions from both the beads of the list and outside the list are calculated. Then, the Coulomb forces are computed

in total in the subsequent timestep ( $n_{step} = 2$ ), Hence, for  $n_{step} > 2$  the Coulomb force acting on a single bead is assessed as sum of the forces derived from the beads in the neighbor list calculated explicitly, and those derived from the outer beads (outside the list) which are calculated by linear extrapolation of the exact forces obtained in the first two timesteps of the multi-step interval. As soon as the condition  $n_{step} = n_{mult}$  is achieved, the procedure is restarted (the neighbor list is again updated). It is readily apparent how this scheme can lead to a significant saving in execution time.

Note that the accuracy of the algorithm is a function of the multiple step interval, and decreases as  $n_{mult}$  increases. Further, the neighbor list removes the need to scan over all the beads in the simulation at every timestep. It is worth to highlight that the algorithm is not time reversible.

We remark that it is also possible to define in input file a maximum displacement  $\Delta r_{pcut}$  in order to achieve a better control on the accuracy of the multiple timestep algorithm. The maximum displacement  $\Delta r_{pcut}$  works as follows: whenever a bead has traveled for a distance larger than  $\Delta r_{pcut}$  from its position at multi-step  $n_{mult} = 1$ , the Verlet neighbor list is updated (even if  $n_{step} < n_{mult}$ ).



# Chapter 3

## JETSPIN Data Files

### Scope of Chapter

This chapter describes all the input and output files of JETSPIN.

### 3.1 Input file

In order to run JETSPIN simulations an input file has to be prepared, which is a free-format and case-insensitive. The input file has to be named `input.dat`, and it contains the selection of the model system, integration scheme directives, specification of various parameters for the model, and output directives. The input file does not require a specific order of key directives, and it is read by the input parsing module. Every line is treated as a command sentence (record). Records beginning with the symbol `#` (commented) and blank lines are not processed, and may be added to aid readability. Each record is read in words (directives and additional keywords and numbers), which are recognized as such by separation by one or more space characters.

As in the example given in `input test`, the last record is a `finish` directive, which marks the end of the input data. Before the `finish` directive, a wide list of directives may be inserted (see Tab 3.1). The key `systype` should be used to set the nanofiber model. In JETSPIN two models are available: 1) the one dimensional model similar to the model of Cap 2 but assuming the nanofiber to be straight along the  $\vec{x}$  axis, and, therefore, neglecting the surface tension force  $\vec{f}_{st}$ ; 2) the three dimensional model described in Cap 2. Internally these options are handled by the integer variable `systype`, which assumes the values explained in Tab 3.1. Further details of the 1-D model can be found in Refs [22, 45]. A series of variables are mandatory and have to be defined. For example, `timestep`, `final time`, `initial length`, etc. (see underlined directives in Tab 3.1). A missed definition of any mandatory variable will call an error banner on the terminal. The user can use two possible ways to define the initial geometry of a nanofiber both given the mandatory directive `initial length`: 1) the discretization step length of nanofiber is specified by the directive `resolution`, which causes automatically the setting of the jet segments number (the number of segment in which the jet is discretised); 2) the jet segments number is declared by the directive `points`, while the value of the discretization step length is automatically set by the program (as in Example **Test Case 3** in Sec 4.3). The directive `cutoff` indicates the length of the upper and lower proximal jet sections, which interact via Coulomb force on any bead. It is worth stressing that the length value is set at the nozzle, so that the effective cutoff increases along the simulation as the nanofiber is stretched. The directive `restart yes` can be used to restart a JETSPIN job (for further information see Section 1.6).

The user should pay special attention in choosing the time integration step given by the directive `timestep`, whose detailed considerations are provided in Ref [20]. The reader is referred to Tab 3.1 for a complete listing of all directives defining the electrospinning setup parameters. Not all these quantities are mandatory, but the user is informed that whenever a quantity is missed, it is usually assumed equal to zero by default (exceptions are stressed in Tab 3.1). **Note all the quantities have to be expressed in centimeter-gram-second unit system** (e.g. charge in statcoulomb, electric potential in statvolt, etc.).

The `job time` and `close time` directives are required to ensure a controlled close down procedure when a job runs out of time. The time specified by the `job time` directive indicates the total time allowed for the job. (This must obviously be set equal to the time specified to the operating system when the job is submitted.) The `close time` directive represents the time JETSPIN will require to write and close all the data files at the end of processing. This means the effective processing time limit is equal to the job time minus the close time.

| directive:                                           | meaning:                                                                                                                                                                                                                                                                               |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>airdrag yes</code>                             | active the inclusion of the airdrag force in the model (default: no)<br>(note that if yes the directive “integrator 4” is mandatory)                                                                                                                                                   |
| <code>airdrag airdensity <math>f</math></code>       | mass density of the air surrounding the nanofiber (default: 0)                                                                                                                                                                                                                         |
| <code>airdrag airviscosity <math>f</math></code>     | kinematic viscosity of the air surrounding the nanofiber (default: 1)                                                                                                                                                                                                                  |
| <code>airdrag airvelocity <math>f</math></code>      | velocity of the air with respect to the $z$ axis (default: 0)                                                                                                                                                                                                                          |
| <code>airdrag amplitude <math>f</math></code>        | amplitude of the randomic force rescaled by $\gamma$ (default: 1)<br>(note that the dissipation coefficient $\gamma$ is automatically computed by JETSPIN using the input data by Eq. 2.15)                                                                                            |
| <code>close time <math>f</math></code>               | Set job closure time to $f$ seconds                                                                                                                                                                                                                                                    |
| <code>collector distance <math>f</math></code>       | distance of collector along the $x$ axis from the nozzle<br>(note the nozzle is assumed in the origin)                                                                                                                                                                                 |
| <code>cutoff <math>f</math></code>                   | length of the proximal jet sections interacting by Coulomb force (default: equal to the collector distance)                                                                                                                                                                            |
| <code>density mass <math>f</math></code>             | mass density of the nanofiber                                                                                                                                                                                                                                                          |
| <code>density charge <math>f</math></code>           | electric volume charge density of the nanofiber                                                                                                                                                                                                                                        |
| <code>dynam refin yes</code>                         | active the dynamic refinement during the simultion                                                                                                                                                                                                                                     |
| <code>dynam refin threshold <math>f</math></code>    | set the threshold beyond which the refinement is applied                                                                                                                                                                                                                               |
| <code>dynam refin every <math>f</math></code>        | the refinement is applied every $f$ seconds                                                                                                                                                                                                                                            |
| <code>dynam refin start <math>f</math></code>        | the refinement is applied only if the simulation time is greater than or equal to $f$ seconds                                                                                                                                                                                          |
| <code>dynam refin anchor <math>f</math></code>       | during the refinement procedure all the anchor beads inserted at distance $f$ preserve their positions<br>(they are mantained as knots of both the old and new meshes)                                                                                                                 |
| <code>evaporation yes</code>                         | activate solvent evaporation and concentration-dependent rheology (default: no)                                                                                                                                                                                                        |
| <code>evaporation polymer frac <math>f</math></code> | initial polymer mass fraction $c_{p,0}$                                                                                                                                                                                                                                                |
| <code>evaporation airviscosity <math>f</math></code> | kinematic viscosity of air $\nu_a$ in $\text{cm}^2/\text{s}$                                                                                                                                                                                                                           |
| <code>evaporation temperature <math>f</math></code>  | temperature in kelvin used for the saturation-vapour correlation                                                                                                                                                                                                                       |
| <code>evaporation umidity <math>f</math></code>      | relative humidity $RH$ in the interval $[0, 1]$<br>(historical spelling retained for input-file compatibility)                                                                                                                                                                         |
| <code>evaporation diffusivity <math>f</math></code>  | solvent-vapour diffusion coefficient $D_a$ in $\text{cm}^2/\text{s}$                                                                                                                                                                                                                   |
| <code>evaporation relumidity <math>f</math></code>   | optional saturation-vapour ratio $c_{s,\text{eq}}(T)$<br>(if omitted it is evaluated from <b>evaporation temperature</b> )                                                                                                                                                             |
| <code>evaporation bconstant <math>f</math></code>    | parameter $B$ in the concentration-dependent viscosity law                                                                                                                                                                                                                             |
| <code>evaporation mconstant <math>f</math></code>    | exponent $m$ in the concentration-dependent viscosity law                                                                                                                                                                                                                              |
| <code>evaporation tconstant <math>f</math></code>    | exponent in the relaxation-time law; use $f = 1$ for Yarin et al. (2001)                                                                                                                                                                                                               |
| <code>elastic modulus <math>f</math></code>          | elastic modulus of the nanofiber                                                                                                                                                                                                                                                       |
| <code>external potential <math>f</math></code>       | electric potential between the nozzle and collector<br>(the resulting electric field is oriented along $x$ -axis)                                                                                                                                                                      |
| <code>external potential freq <math>f</math></code>  | frequency of the waveform describing the electric potential                                                                                                                                                                                                                            |
| <code>external potential phase <math>f</math></code> | initial phase of the waveform describing the electric potential                                                                                                                                                                                                                        |
| <code>external potential type <math>i</math></code>  | set the waweform describing the time evolution of the external electric potential. The <i>integer</i> can be equal to 0 for a static field,<br>1 for a rectangular waveform,<br>2 for a rectangular waveform of a RC circuit,<br>3 for orthogonal rotating electric field (default: 0) |
| <code>external potential time <math>f</math></code>  | time constant (relaxation time) of the RC circuit                                                                                                                                                                                                                                      |

|                                |                                                                                                                                                                                                            |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| external potential vector $3f$ | vector of the electric field for $x y z$ components                                                                                                                                                        |
| <u>final time</u> $f$          | set the end time of the simulation                                                                                                                                                                         |
| <u>finish</u>                  | close the input file (last data record)                                                                                                                                                                    |
| gravity yes                    | active the inclusion of the gravity force in the model (default: no)                                                                                                                                       |
| hbfluid yes                    | allow the set of parameters needed to define the Herschel–Bulkley stress. Note that if it is set no, JETSPIN will use the simple Maxwellian fluid model (default: no)                                      |
| hbfluid consistency $f$        | flow consistency index rescaled by $\mu$ (in $\mu$ unit) as defined in Eq 2.5 (Note you should also set: hbfluid yes) (default: 1)                                                                         |
| hbfluid index $f$              | flow behavior index (Note you should also set: hbfluid yes) (default: 1)                                                                                                                                   |
| <u>initial length</u> $f$      | length of the nanofiber at the initial time                                                                                                                                                                |
| <u>integrator</u> $i$          | set the integration scheme. The <i>integer</i> can be '1' for Euler, '2' for Heun, '3' for Runge-Kutta, and '4' for the Platen scheme (only if airdrag force is activated).                                |
| inserting yes                  | inject new beads as explained in Subsec                                                                                                                                                                    |
| job time $f$                   | Set job time to $f$ seconds                                                                                                                                                                                |
| kvfluid yes                    | activate the Kelvin–Voigt model which is used instead of Maxwell model (default: no) (Note it is actually implemented only for working with the 3d-model without airdrag activated: system 3 is mandatory) |
| maximum displ $f$              | set the maximum displacement of the neighbor list (see Sec 2.10)                                                                                                                                           |
| multiple step yes              | activate the multiple timestep algorithm for Coulomb forces (default: no)                                                                                                                                  |
| multiple step every $i$        | set the interval of multiple-steps (see Sec 2.10)                                                                                                                                                          |
| noise yes                      | active the inclusion of a random noise in the model (default: no) (note that if yes the directive “integrator 4” is mandatory)                                                                             |
| noise variance $f$             | variance of random noise in velocity space (default: $1 \text{ cm}^2/\text{s}^2$ )                                                                                                                         |
| noise diffusivity $f$          | diffusivity of random noise in velocity space (default: $0 \text{ cm}^2/\text{s}^3$ )                                                                                                                      |
| <u>nozzle cross</u> $f$        | cross section radius of the nanofiber at the nozzle                                                                                                                                                        |
| nozzle stress $f$              | viscoelastic stress of the nanofiber at the nozzle                                                                                                                                                         |
| nozzle velocity $f$            | bulk fluid velocity of the nanofiber at the nozzle                                                                                                                                                         |
| perturb yes                    | active the periodic perturbation at the nozzle                                                                                                                                                             |
| perturb freq $f$               | frequency of the perturbation at the nozzle                                                                                                                                                                |
| perturb ampl $f$               | amplitude of the perturbation at the nozzle                                                                                                                                                                |
| points $i$                     | number of segment in which the jet is discretised                                                                                                                                                          |
| primary cutoff $f$             | set the primary cutoff of the neighbor list (see Sec 2.10)                                                                                                                                                 |
| print list $s_1 \dots$         | print on terminal statistical data as indicated by symbolic strings (see Tab 3.2 for detailed informations)                                                                                                |
| print time $f$                 | print data on terminal and output file every $f$ seconds                                                                                                                                                   |
| print pdb frame $f$            | print the jet geometry in a single PDB file format in centimeters                                                                                                                                          |
| print pdb rescalepdb $f$       | print the PDB file format data rescaled by $f$ (default: 1)                                                                                                                                                |
| print xyz $f$                  | print the trajectory in XYZ file format in centimeters every $f$ seconds                                                                                                                                   |
| print xyz frame $f$            | print the jet geometry in a single XYZ file format in centimeters every $f$ seconds (default name of files: frame%06d.xyz)                                                                                 |
| print xyz maxnum $i$           | print the XYZ file format with $i$ beads starting from the nearest bead to the collector (default: 100)                                                                                                    |
| print xyz rescale $f$          | print the XYZ file format data rescaled by $f$ (default: 1)                                                                                                                                                |
| printstat list $s_1 \dots$     | print on output file statistical data as indicated by symbolic strings (see Tab 3.2 for detailed informations)                                                                                             |
| removing yes                   | remove beads at collector                                                                                                                                                                                  |
| restart yes                    | restart JETSPIN from a previous job (see Section 1.6)                                                                                                                                                      |
| restart dump $i$               | print the <b>save.dat</b> file every $i$ steps                                                                                                                                                             |
| restart reset                  | reset all the accumulated statistical data collected in the previous simulation when a job is restarted                                                                                                    |
| resolution $f$                 | discretization step length of nanofiber                                                                                                                                                                    |

|                     |                                                                                                                                                                     |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| seed $i$            | seed control to the internal random number generator                                                                                                                |
| surface tension $f$ | surface tension coefficient of the nanofiber                                                                                                                        |
| <u>system</u> $i$   | set the nanofiber model. The <i>integer</i> can be equal to<br>'1' for select the 1d-model, '3' for the 3d-model, and<br>'4' for the 3d-model with airdrag ativated |
| timestep $f$        | set the time step for the integration scheme                                                                                                                        |
| viscosity $f$       | viscosity of the nanofiber                                                                                                                                          |
| yield stress $f$    | yield stress (default: 0)                                                                                                                                           |

Table 3.1: Here, we report the list of directives available in JETSPIN. Note  $i$ ,  $f$ , and  $s$  denote an integer number, a floating point number, and a string, respectively. The underlined directives are mandatory. The default value is zero (exceptions are stressed in parentheses).

### 3.1.1 Evaporation directives

The evaporation directives listed in Table 3.1 activate and parameterize the model described in Section 2.8:

**evaporation yes** activate solvent evaporation and the concentration-dependent rheology.

**evaporation polymer frac**  $f$  initial polymer mass fraction  $c_{p,0}$ .

**evaporation airviscosity**  $f$  kinematic viscosity of air  $\nu_a$  in  $\text{cm}^2/\text{s}$ .

**evaporation temperature**  $f$  temperature in kelvin.

**evaporation umidity**  $f$  relative humidity  $RH$  in the interval  $[0,1]$ . The historical spelling **umidity** is retained for input-file compatibility.

**evaporation diffusivity**  $f$  vapour diffusion coefficient  $D_a$  in  $\text{cm}^2/\text{s}$ . If omitted, Eq. 2.37 is used at one atmosphere.

**evaporation relumidity**  $f$  optional value of the saturation-vapour ratio  $c_{s,\text{eq}}(T)$ . If omitted, it is evaluated from **evaporation temperature**. The historical keyword is retained for input-file compatibility.

**evaporation bconstant**  $f$  parameter  $B$  in Eq. 2.45.

**evaporation mconstant**  $f$  exponent  $m$  in Eq. 2.45.

**evaporation tconstant**  $f$  exponent  $t_{\text{ev}}$  in the generalized relaxation-time law; use  $f = 1$  to reproduce Yarin et al. (2001).

## 3.2 Output files

A series of specific directives causes the writing of output files (see Tab 3.1). In JETSPIN three output files can be written: 1) the file **statdat.dat** containing time-dependent statistical data of simulated process; 2) the file **traj.xyz** reporting the nanofiber trajectory in XYZ format file; 3) the files **frame%06d.xyz** reporting the nanofiber geometries in splitted XYZ format files (a single file for each frame); 4) the files **frame%06d.pdb** reporting the nanofiber geometries in splitted PDB (Protein Data Bank) format files (a single file for each frame). Note that in the last case a topological file **frame%06d.psf** in PSF (Protein Structure File) format will be printed alongside each pdb file. These two files (PDB and PSF) can be opened by VMD-Visual Molecular Dyanamics [46].

Various statistical data can be written on the file **statdat.dat**, and the user can select them in input using the directive **printstat list** followed by appropriate symbolic strings, whose the list with corresponding meanings is reported in Tab 3.2. The selected observables will be printed at time interval on the same line

following the order specified in input file. In the same way, a list of statistical data can be printed on computer terminal using the directive `print list` followed by the symbolic strings of Tab 3.2.

At normal termination, the terminal also reports the wall-clock duration of the temporal integration loop and its throughput in integration steps per second. The timer starts immediately before the loop and stops immediately after it; input parsing, initial allocation, restart loading, and output-file setup are therefore excluded. Integration, bead insertion and removal, statistics, and scheduled output executed inside the loop are included. In MPI builds, ranks are synchronized at both boundaries and rank zero reports the elapsed duration of the slowest participating rank. This loop-only wall time is the recommended measurement for CPU, MPI, and accelerator performance comparisons.

For performance diagnosis, setting the environment variable `JETSPIN_PROFILE=1` prints an optional wall-clock breakdown for the integrator, Coulomb calculation, bead topology operations, statistics, scheduled output, and restart output. Profiling is disabled by default. The Coulomb value is a nested part of the integrator total and must not be added to it when interpreting percentages. The profiler prints cumulative wall time, percentage of the complete temporal-loop time, and call count on the terminal; it does not write a profiling file. These are explicitly instrumented regions rather than an automatic measurement of every Fortran subroutine. Region timings are inclusive, so the integrator value includes the Coulomb, equation-of-motion evaluation, and Runge–Kutta update regions.

In MPI execution, the detailed region breakdown is local to rank zero and is not reduced to the maximum over all ranks. The synchronized complete-loop wall time should be used for MPI scaling. Since clock reads add a small overhead, performance comparisons should enable profiling in both runs or disable it in both runs.

The file `traj.xyz` is written as a continuous series of XYZ format frames taken at time interval, so that can be read by suitable visualization programs (e.g. VMD-Visual Molecular Dynamics [46], UCSF Chimera [47], etc.) to generate animations. The number of elements contained in the file is kept constant equal to the value specified by the directive `print xyz maxnum`, since few programs (e.g. VMD) do not manage a variable number of elements along the simulations. If the actual number of beads is lower than the given constant `maxnum`, the extra elements are printed in the origin point. On the other hand, the number of beads printed in the files `frame%06d.xyz` and `frame%06d.pdb` is exactly the actual number of beads used to discretized the jet at the given frame. Note that the unit of measurement used in the files, `traj.xyz`, `frame%06d.xyz` and `frame%06d.pdb` is the centimeter multiplied by  $f$ , which was given in input file by the directive “`print xyz rescale f`” and “`print pdb rescalepdb f`” (see Table 3.1), respectively.

| keys:         | meaning:                                                     |
|---------------|--------------------------------------------------------------|
| t             | unscaled time                                                |
| ts            | scaled time                                                  |
| x, y, z       | unscaled coordinates of the fareset bead                     |
| xs, ys, zs    | scaled coordinates of the fareset bead                       |
| st            | unscaled stress of the fareset bead                          |
| sts           | scaled stress of the fareset bead                            |
| vx, vy, vz    | unscaled velocities of the fareset bead                      |
| vxs, vys, vzs | scaled velocities of the fareset bead                        |
| yz            | unscaled normal distance from the x axis of the fareset bead |
| yzs           | scaled normal distance from the x axis of the fareset bead   |
| mass          | last inserted mass at the nozzle                             |
| q             | last inserted charge at the nozzle                           |
| cpu           | time for every print interval                                |
| cpur          | remaining time to the end                                    |
| cpue          | elapsed time                                                 |
| n             | number of beads used to discretise the jet                   |
| f             | index of the first bead                                      |
| l             | index of the last bead                                       |
| <u>curn</u>   | current at the nozzle                                        |
| <u>curc</u>   | current at the collector                                     |
| <u>vn</u>     | velocity modulus of jet at the nozzle                        |

|            |                                                      |
|------------|------------------------------------------------------|
| <u>vc</u>  | velocity modulus of jet at the collector             |
| <u>svc</u> | strain velocity at the collector                     |
| <u>mfn</u> | mass flux at the nozzle                              |
| <u>mfc</u> | mass flux at the collector                           |
| <u>rc</u>  | radius of jet at the collector                       |
| <u>rrr</u> | radius reduction ratio of jet                        |
| <u>lp</u>  | length path of jet                                   |
| <u>rlp</u> | length path of jet divided by the collector distance |
| mxst       | maximum stress along the jet                         |
| mxsx       | x position of the jet maximum stress                 |

---

Table 3.2: In JETSPIN a series of instantaneous and statistical data are available to be printed by selecting the appropriate key. Here, the list of symbolic string keys is reported with their corresponding meanings. Note by *scaled* we mean that the observable was rescaled by the characteristic values provided in Tab2.1. The underlined keys correspond to data which are averaged on the time interval given by the directive **print time** in input file. Note by *farest bead* we mean the farest bead from the nozzle (with greatest  $x$  value). Note all the quantities are expressed in centimeter-gram-second unit system, excepted the dimensionless scaled observables.

## Chapter 4

# Example Simulations

### Scope of Chapter

This chapter describes the standard test cases for JETSPIN, the input and output files for which are in the *examples* sub-directory.

#### 4.1 Test Case 1: 1-D case

This input file (located in the *input-1* sub-directory) provides the dimensionless parameter values  $Q = 12$ ,  $V = 2$  and  $F_{ve} = 12$ , which have been already used as reference case in Refs [11, 22].

#### 4.2 Test Case 2: 1-D case with bead insertion at the nozzle activated

This input file (located in the *input-2* sub-directory) provides the dimensionless parameter values  $Q = 12$ ,  $V = 2$  and  $F_{ve} = 12$ , as in the previous test case, but here we have activated the injection of new beads by the directive `inserting yes` in the input file.

#### 4.3 Test Case 3: Three-D simulation of electrospinning process

The electrospinning of polyvinylpyrrolidone (PVP) nanofibers is a prototypical process, which has been largely investigated in literature.[6, 8, 5] By the input located in the *input-3* sub-directory, we simulate the electrospinning process of PVP solutions. Then, the theoretical results predicted by the models are compared with the aforementioned experimental data. In particular, we reproduce an experiment in which a solution of PVP (molecular weight = 1300 kDa) is prepared by a mixture of ethanol and water (17:3 v:v), at a concentration of about 2.5 wt%. The applied voltage is in a range around 10 kV, and the collector is placed at distance 16 cm from the nozzle, which has radius 250 micron (further details are provided in Ref. [48, 20]). As rheological properties of such system we consider the zero-shear viscosity  $\mu_0 = 0.2 \text{ g}/(\text{cm} \cdot \text{s})$  [49, 50], the elastic modulus  $G = 5 \cdot 10^4 \text{ g}/(\text{cm} \cdot \text{s}^2)$  [51], and the surface tension  $\alpha = 21.1 \text{ g}/\text{s}^2$  [49]. We use for the simulation a viscosity value  $\mu$  which is two order of magnitude larger than the zero-shear viscosity  $\mu_0$  reported, since the strong longitudinal flows we are dealing with can lead to an increase of the extensional viscosity from  $\mu_0$ , as already observed in literature.[11, 32] The mass density is equal to  $0.84 \text{ g}/\text{cm}^3$ , while the charge density was estimated by experimental observations of the current measured at the nozzle. For convenience, all the simulation parameters are summarized in Tab 4.1.



| $\rho$<br>(kg/m <sup>3</sup> ) | $\rho_q$<br>(C/L)   | $a_0$<br>(cm)     | $v_s$<br>(cm/s) | $\alpha$<br>(mN/m) | $\mu$<br>(Pa·s) | $G$<br>(Pa) | $V_0$<br>(kV) | $\omega$<br>(s <sup>-1</sup> ) | $A$<br>(cm) |
|--------------------------------|---------------------|-------------------|-----------------|--------------------|-----------------|-------------|---------------|--------------------------------|-------------|
| 840                            | $2.8 \cdot 10^{-7}$ | $5 \cdot 10^{-3}$ | 0.28            | 21.1               | 2.0             | 50000       | 9.0           | $10^4$                         | $10^{-3}$   |

Table 4.1: Simulation parameters for the simulation of PVP nanofibers. The headings used are as follows:  $\rho$ : density,  $\rho_q$ : charge density,  $a_0$ : fiber radius at the nozzle,  $v_s$ : bulk fluid velocity in the syringe needle,  $\alpha$ : surface tension,  $\mu$ : viscosity,  $G$ : elastic modulus,  $V_0$ : applied voltage bias,  $\omega$ : frequency of perturbation,  $A$ : amplitude of perturbation. The bulk fluid velocity  $v_s$  was estimated considering that the solution was pumped at constant flow rate of 2 mL/h in a needle of radius 250 micron.

#### 4.4 Test Case 4: Three-D simulation in presence of a gas counter-flow

#### 4.5 Test Case 5: Three-D simulation with dynamic refinement

#### 4.6 Test Case 6: Three-D simulation of a quasi-Newtonian fluid

#### 4.7 Test Case 7: Three-D simulation with evaporation and an orthogonal rotating electric field

Test Case 7 combines the three-dimensional Maxwell model with an orthogonal rotating electric field and the solvent evaporation and solidification model described in Sec. 2.8. The setting `evaporation tconstant 1` reproduces the relaxation-time law of Eq. (38) of Yarin, Koombhongse and Reneker [38]. The evaporation cutoff is the Yarin criterion  $c_s = 0.1$ , which in JETSPIN is imposed through the equivalent volume ratio  $V_i/V_i^0 = c_{p,0}/0.9$ .

The example in `examples/input-7/input.dat` is intended as a regression and reference case for the coupled JETSPIN implementation. Its initial polymer fraction is  $c_{p,0} = 0.025$  and therefore it should not be interpreted as a quantitative reproduction of the 6% aqueous-PEO case used in the original Yarin et al. comparison. The orthogonal rotating field is a separate JETSPIN extension and is not part of the 2001 evaporation model.

#### 4.8 Test Case 8: Yarin et al. (2001) reference parameters

The directory `examples/input-8` contains a dedicated reference case based on the 6 wt% aqueous PEO calculations of Ref. [38]. The values used there are  $a_0 = 0.015$  cm,  $\rho = 1$  g/cm<sup>3</sup>,  $\alpha = 70$  dyn/cm,  $\mu_0 = 10^4$  P,  $\theta_0 = 10^{-2}$  s, a charge density corresponding to 1 C/L,  $h = 20$  cm, and an applied field of 1.5 kV/m. In JETSPIN cgs units these correspond to  $G_0 = \mu_0/\theta_0 = 10^6$  Ba, a charge density  $2.99792458 \times 10^6$  statC/cm<sup>3</sup>, and an applied potential of approximately 1 statV across the 20 cm gap. The ambient conditions are  $T = 293.15$  K and  $RH = 0.165$ , while the solidification parameters are  $B = 7$ ,  $m = 0.1$  and  $t_{ev} = 1$ .

The paper reports  $K_s = 100$ . Since  $K_s = \omega\theta_0$ , Test Case 8 uses  $\omega = 10^4$  s<sup>-1</sup>. The small perturbation amplitude used to seed bending is a JETSPIN numerical choice rather than a value specified in the dimensional parameter list of Ref. [38]. The initial bead spacing is similarly chosen for compatibility with the JETSPIN injection algorithm: the test uses  $l_{step} = 0.0325$  cm, the electrostatic cutoff length estimated in the same paper. Therefore Test Case 8 is intended as an order-of-magnitude reference and regression test of the evaporation/rheology implementation, rather than as a pointwise reproduction of the original discretized trajectory.

The test intentionally omits the `evaporation diffusivity` directive, so that Eq. 2.37 is exercised. At 293.15 K the expected value is  $D_a \approx 0.242001758$  cm<sup>2</sup>/s. The solidification cutoff is

$$\frac{V_i}{V_i^0} = \frac{0.06}{0.9} = 0.0666666667, \quad (4.1)$$



for which  $c_p = 0.9$ . With  $B = 7$  and  $m = 0.1$ , the corresponding reference ratios are

$$\frac{\mu}{\mu_0} \simeq 43.9783350, \quad \frac{\theta}{\theta_0} = 15, \quad \frac{G}{G_0} \simeq 2.9318890. \quad (4.2)$$

These values provide simple numerical targets for future regression checks of the evaporation cutoff and concentration-dependent rheology.

## 4.9 Test Case 9: 1,000-bead CPU/GPU performance benchmark

Test Case 9 is a performance-oriented three-dimensional Maxwell simulation derived from Test Case 7. It initializes 1,000 beads uniformly over the full 16 cm distance from the nozzle to the collector and executes 1,000 integration steps. Insertion, removal, multiple-step Coulomb summation, and dynamic refinement are disabled, so the workload remains fixed.

The material, aerodynamic, gravity, and perturbation parameters are retained from Test Case 7. The orthogonal rotating field is replaced by a constant axial electric potential. Evaporation is disabled because its direct Coulomb routine is not yet accelerated by the initial OpenACC implementation. The case therefore exercises the direct  $O(N^2)$  Coulomb kernel without an evaporation fallback or topology changes.

Test Case 9 is intended for performance comparison rather than as an additional physical validation baseline. CPU and accelerator runs should be compared using the terminal value `Time-integration loop wall time`, which excludes input processing, initial allocation, and accelerator initialization. Its input is stored in `examples/input-9/input.dat`.

After compiling separate CPU and accelerator executables, copy each executable and `examples/input-9/input.dat` into a separate run directory. Execute the two cases sequentially on an otherwise idle system and compare their `Time-integration loop wall time` values. The compiler version, GPU model, GPU compute capability, and build options should accompany every reported benchmark.

An initial reference measurement with NVFORTRAN 24.3 and one NVIDIA A30 reported 40.659856 s for the CPU executable and 3.785055 s for the OpenACC executable, corresponding to a speedup of 10.742. Both runs retained exactly 1,000 beads and agreed within the numerical regression tolerances. These timings are indicative and should be remeasured on each target system.

An optional internal subroutine profiler is enabled at run time with `JETSPIN_PROFILE=1 ./main.x`; no rebuild is required. The report separates the integrator, Coulomb force, bead topology operations, statistics, scheduled output, and restart output. The Coulomb region is nested within the integrator total and the two values must not be added.

The report is written only to the terminal and gives cumulative wall time, percentage of the complete temporal loop, and call count for each explicitly instrumented region. It is not an automatic profile of every Fortran subroutine. The fourth-order Runge–Kutta integration used by this test makes four Coulomb evaluations per step, yielding 4,000 calls. Clock reads add a small overhead, so profiling must have the same enabled or disabled state in both runs being compared. In MPI execution, the detailed table is local to rank zero rather than a maximum over ranks; the synchronized complete-loop timer remains the appropriate MPI scaling measurement.

In an initial profiled run, the NVFORTRAN CPU loop required 39.813621 s, of which 38.232985 s (96.03%) was spent in 4,000 Coulomb evaluations. The A30 loop required 3.320607 s, with 2.164045 s (65.17%) in the same Coulomb region. The corresponding speedups were approximately 11.99 for the complete loop and 17.67 for Coulomb alone. Repeating this profile after each porting milestone provides a direct view of time moving from the Coulomb kernel to remaining host work, device transfers, or other force terms.

The complete sampled CPU and A30 OpenACC numerical outputs from this initial measurement are versioned in `tests/performance/test9/`. The accompanying comparison script checks later CPU and GPU results with relative tolerance  $10^{-6}$  and absolute tolerance  $10^{-9}$ . These records remain outside the automatic regression matrix because of the benchmark's longer execution time.

A subsequent profiler refinement measured 2.026551 s (56.281420609 s (39.45(0.41equation-of-motion and Coulomb region was called 4,000 times. This breakdown selects the three-dimensional equation-of-motion chain as the next meaningful accelerator target; porting the vector updates alone cannot materially improve this workload. The trajectory still matched the versioned A30 reference exactly at the saved output precision.

The first equation-of-motion offload milestone uses one explicit OpenACC kernel per RK4 stage for the Test 9 force assembly. The local three-point curvature construction also executes entirely on the device: every bead reads only itself and its two neighbours, and no curvature array is transferred from the host. Because the initially straight jet makes the circumcentre arithmetic sensitive to floating-point contraction, the OpenACC target uses the NVFORTRAN `nofma` GPU option. In an updated paired measurement built from the same source with NVFORTRAN 24.3, the CPU and A30 complete-loop times were 38.616119 s and 3.159748 s, respectively, for a speedup of 12.22. The CPU/A30 regional times were 37.059132/2.250507 s for Coulomb (16.47 speedup), 1.398234/0.736914 s for equation-of-motion evaluation including curvature (1.90 speedup), and 0.018556/0.030343 s for the Runge–Kutta updates. Coulomb and equation-of-motion evaluation each had 4,000 calls. The slower device-side RK updates represented less than one percent of the A30 loop. The complete saved trajectory passed the A30 baseline comparison with relative tolerance  $10^{-6}$  and absolute tolerance  $10^{-9}$ ; the worst normalized difference was  $7.82 \cdot 10^{-5}$ . That first EOM milestone used call-scoped transfers and was restricted to the fixed Test 9 configuration; other configurations retain the CPU path.

The subsequent persistent-data milestone maps the complete Test 9 state, Coulomb force, equation-of-motion derivatives, and Runge–Kutta scratch arrays once and retains them on the A30 across timesteps. Cross sections and all four Runge–Kutta updates are computed on the device, and the Coulomb result is consumed directly by the equation-of-motion kernel. The existing CPU statistics still require one update of the seven primary state arrays after each timestep, corresponding to 56,056 bytes per step for 1,001 beads.

In a representative profiled run, the persistent complete loop took 2.451978 s, compared with 3.159748 s for the call-scoped A30 implementation and 38.616119 s for the paired NVFORTRAN CPU run. The persistent Coulomb, equation-of-motion, and Runge–Kutta regions took 2.006970 s, 0.112892 s, and 0.191860 s, respectively. The Runge–Kutta region includes the once-per-step full-state update to the host. The complete-loop speedup over the CPU was 15.75. The trajectory passed the unchanged A30 baseline with relative tolerance  $10^{-6}$  and absolute tolerance  $10^{-9}$ ; its worst normalized difference was  $7.8 \cdot 10^{-5}$ .

In the following device-resident-statistics milestone, the path length and maximum stress are accumulated on the accelerator after each step. Their scalar counters remain on the device. The complete primary state is copied to the host only at the five scheduled 200-step samples. The final restart records the last synchronized timestep and reuses the step-1000 host snapshot rather than downloading it again. Thus only five full-state synchronization events remain. A representative A30 run took 2.468580 s for the complete loop, of which 0.056676 s was attributed to the small statistic kernels. The trajectory passed the unchanged A30 baseline with relative tolerance  $10^{-6}$  and absolute tolerance  $10^{-9}$ ; the worst normalized difference was  $7.97 \cdot 10^{-5}$ .

The next milestone fuses the path-length and maximum-stress reductions with the final Runge–Kutta state update. Each device thread reconstructs the updated position of the next bead from the same Runge–Kutta coefficients, avoiding a neighbour race. Two small follow-up kernels preserve the rule that the last bead index wins when maximum stresses are equal. In a representative A30 run, the statistics region decreased from 0.056676 s to 0.036373 s, and the complete loop took 2.457896 s. An extended comparison matched maximum stress and its position exactly; path length differed by at most  $2 \cdot 10^{-8}$  cm because of reduction ordering.

# Chapter 5

## Tips

### Scope of Chapter

This chapter describes few useful tips which can be used to enhance results obtained by JETSPIN.

#### 5.1 How to visualize a PDB file in VMD software package

The PDB file written by JETSPIN can be easily visualized in VMD software package. The first step is to download and install VMD which can be found at website: <http://www.ks.uiuc.edu/Research/vmd/>. As VMD is installed, the PDB file can be opened using VMD: "vmd frame000001.pdb" in the terminal or click on "file" in the main white box and then on "load molecule" (see Fig 5.1), browse for your PDB file frame000001.pdb and load it. In order to load the topological structure of the jet, it is possible to load the PSF file written by JETSPIN alongside the PDF file: right click on the just uploaded file (e.g. "frame000001.pdb") in the white box in the middle of the "MD Main" window, and select "Load Coordinates into Molecule" pull-down menu option (see Fig 5.2). Hence, browse for the PSF file (frame000001.psf) defining the bonds for each pair of consecutive jet beads, and load it. Finally, the jet will be displayed in the graphics window.

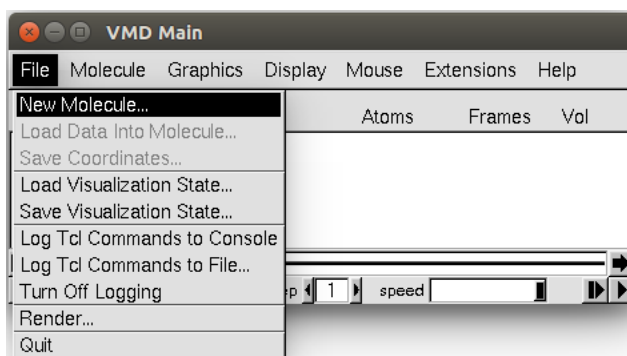


Figure 5.1: Snapshot of the main white box of VMD.

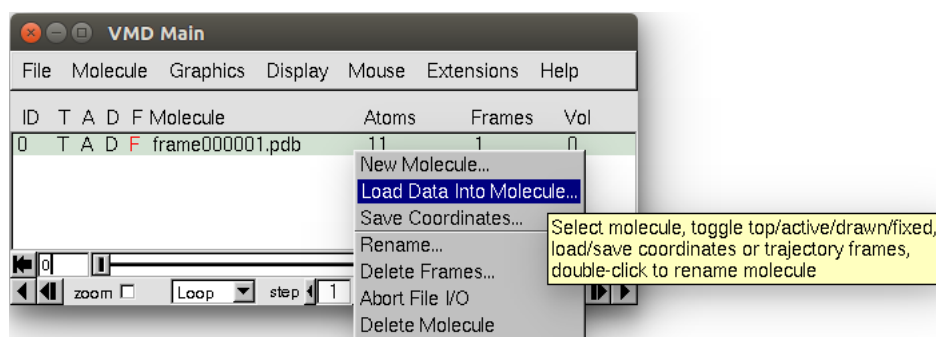


Figure 5.2: Snapshot of the "Load Coordinates into Molecule" selection in VMD.

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